ERS statement on benign pleural effusions in adults

Anand Sundaralingam[1,2,3], Elzbieta M. Grabczak[4], Patrizia Burra[5], M. Inês Costa[6], Vineeth George[7,8], Eli Harriss[9], Ewa A. Jankowska[10], Julius P. Janssen[11], Georgia Karpathiou[12], Christian B. Laursen[13,14], Kornelija Maceviciute[15], Nick Maskell[16], Federico Mei[17,18], Blin Nagavci[19], Vasiliki Panou[13,14,20], Valentina Pinelli[21], José M. Porcel[22], Sara Ricciardi[23,24], Samira Shojaee[25], Hugh Welch[16], Alberto Zanetto[5], Udaya Prabhakar Udayaraj[26,27], Giuseppe Cardillo[28,29] and Najib M. Rahman[1,2,30,31]

[1]Oxford Respiratory Trials Unit, Churchill Hospital, Headington, UK. [2]Oxford Pleural Unit, Oxford Centre for Respiratory Medicine, Oxford University Hospitals NHS Trust, Oxford, UK. [3]School of Pharmacy and Biomedical Sciences, University of Portsmouth, Portsmouth, UK. [4]Department of Internal Medicine, Pulmonary Diseases and Allergy, Medical University of Warsaw, Warsaw, Poland. [5]Gastroenterology and Multivisceral Transplant Unit, Department of Surgery, Oncology, and Gastroenterology, Padua University Hospital, Padua, Italy. [6]Pulmonology Department, Centro Hospitalar Universitário de Santo António, Porto, Portugal. [7]Department of Respiratory and Sleep Medicine, John Hunter Hospital, Newcastle, Australia. [8]Hunter Medical Research Institute, Newcastle, Australia. [9]Bodleian Health Care Libraries, University of Oxford, Oxford, UK. [10]Division of Translational Cardiology and Clinical Registries, Institute of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland. [11]Dept of Pulmonary Diseases, Canisius Wilhelmina Hospital, Nijmegen, The Netherlands. [12]Pathology Department, University Hospital of Saint Etienne, Saint Etienne, France. [13]Department of Respiratory Medicine, Odense University Hospital, Odense, Denmark. [14]Odense Respiratory Research Unit (ODIN), Department of Clinical Research, University of Southern Denmark, Odense, Denmark. [15]St Bartholomew's Hospital, London, UK. [16]Academic Respiratory Unit, University of Bristol, Bristol, UK. [17]Department of Biomedical Sciences and Public Health, Polytechnic University of Marche, Marche, Italy. [18]Respiratory Disease Unit, University Hospital, Ancona, Italy. [19]Institute for Evidence in Medicine, Faculty of Medicine and Medical Center, University of Freiburg, Freiburg, Germany. [20]Department of Respiratory Diseases, Aalborg University Hospital, Aalborg, Denmark. [21]Division of Pulmonology, ASL5 Spezzino, La Spezia, Italy. [22]Pleural Medicine Unit, Arnau de Vilanova University Hospital, Lleida, Spain. [23]Division of Thoracic Surgery, San Camillo Forlanini Hospital, Rome, Italy. [24]PhD program Alma Mater Studiorum, University of Bologna, Bologna, Italy. [25]Division of Allergy, Pulmonary and Critical Care Medicine, Vanderbilt University Medical Center, Nashville, TN, USA. [26]Oxford Kidney Unit, Churchill Hospital, Oxford, UK. [27]Nuffield Department of Medicine, Henry Wellcome Building for Molecular Physiology, University of Oxford, Oxford, UK. [28]Unit of Thoracic Surgery, Azienda Ospedaliera San Camillo Forlanini, Rome, Italy. [29]Unicamillus, International University of Health Sciences, Rome, Italy. [30]NIHR Oxford Biomedical Research Centre, University of Oxford, Oxford, UK. [31]Chinese Academy of Medical Health Sciences, University of Oxford, Oxford, UK.

Corresponding author: Anand Sundaralingam (Anand.Sundaralingam@ndm.ox.ac.uk)

Shareable abstract (@ERSpublications)
The management of benign pleural effusions can be complex, often existing at the junction of multiple specialty groups. Multidisciplinary working and a focus on patient priorities are essential in delivering good care for these patients. https://bit.ly/3y1fHB5

Cite this article as: Sundaralingam A, Grabczak EM, Burra P, *et al*. ERS statement on benign pleural effusions in adults. *Eur Respir J* 2024; 64: 2302307 [DOI: 10.1183/13993003.02307-2023].

Copyright ©The authors 2024. For reproduction rights and permissions contact permissions@ersnet.org

Received: 22 Dec 2023
Accepted: 10 June 2024

Abstract

The incidence of non-malignant pleural effusions far outweighs that of malignant pleural effusions and is estimated to be at least 3-fold higher. These so-called benign effusions do not follow a "benign course" in many cases, with mortality rates matching and sometimes exceeding those of malignant pleural effusions. In addition to the impact on patients, healthcare systems are also significantly affected, with recent US epidemiological data demonstrating that 75% of resource allocation for pleural effusion management is spent on non-malignant pleural effusions (excluding empyema). Despite this significant burden of disease, and by existing at the junction of multiple medical specialties, reflecting a heterogenous constellation of medical conditions, non-malignant pleural effusions are rarely the focus of research or the subject of management guidelines. With this European Respiratory Society Task Force, we assembled a multispecialty collaborative across 11 countries and three continents to provide a statement based on systematic searches of the medical literature to highlight evidence in the management of the following clinical areas: a diagnostic approach to transudative effusions, heart failure, hepatic hydrothorax, end-stage renal failure, benign asbestos-related pleural effusion, post-surgical effusion and nonspecific pleuritis.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Introduction

Pleural effusions are a common medical presentation, with an estimated incidence of 337 per 100 000 of the population [1]. Of these, the majority of cases are due to non-malignant pleural effusions (NMPEs), also known as "benign" pleural effusions (252 per 100 000) [2]. Despite their prevalence, there is an absence of established guidelines to support clinicians in diagnosis and management. In part, this may be explained by a lack of high-quality evidence; however, the heterogeneity of conditions that lead to NMPE, poor definitions and classification systems, and patient care spread across different speciality groups has almost certainly contributed to this paucity of guidance [3]. Nonetheless, such patients carry a high symptom burden and significant morbidity and mortality. Observational series have estimated high 1-year mortality rates in heart (50%), renal (46%) and liver (25%) failure, rendering the term "benign" something of a misnomer [4]. The impact is felt by healthcare systems across the world. Recent epidemiological data from the USA suggest 75% of the total resource allocation for pleural effusions was spent on the management of NMPE (excluding empyema) alone [5]. With an ageing population and medical advances, the proportion of patients living with chronic disease is rising. Thus, the incidence of NMPE is expected to increase [1]. It is therefore imperative to support clinicians who routinely manage the care of such patients.

This statement aims to produce a narrative review of the current evidence with regards to the management of NMPE in adults. The statement specifically excludes the management of pleural infection, which, in contrast to the other forms of "benign" pleural effusion, is supported by high-quality evidence and a number of international guidelines and consensus statements [6]. The Task Force (TF) considered other topics that may be considered part of the spectrum of "benign" pleural effusions (chronic lymphocytic exudates, connective tissue-related effusions and chylothorax) but ultimately settled on the groups presented below in view of their respective clinical burden of disease, paucity of existing guidance and suitability for future standalone guidance in the topics specifically excluded.

The current statement does not make clinical practice recommendations. However, in specific areas where the evidence is scarce or mixed, such limitations are described and a description of the TF members' own experience and practice is provided for information, but not with the aim of guiding clinical practice.

Methods

A TF was formed with the goal of producing a statement: a comprehensive, scientific review of the literature, identified by systematic searches, with conclusions supported by accompanying references. Membership comprised nine respiratory physicians (with subspecialist expertise in pleural disease and interventional pulmonology), one thoracic surgeon, two hepatologists, one nephrologist and one cardiologist, with the support of five early-career members. There was representation from nine European countries, the USA and Australia.

The scope of the statement was agreed at the initial meeting in January 2022, specifically outlining the topics an approach to transudative effusions, heart failure (HF), hepatic hydrothorax (HH), end-stage renal failure (ESRF), benign asbestos-related pleural effusion, post-surgical pleural effusion and nonspecific pleuritis (NSP). Working groups produced key questions in a Population, Intervention, Comparator and Outcomes (PICO) style format for each topic, assisted by a European Respiratory Society (ERS) methodologist (B. Nagavci). Following TF debate, a number of questions were selected for each group. The literature search was undertaken for each PICO question by each working group with the assistance of a medical librarian (E. Harriss). MEDLINE (National Library of Medicine, USA), Ovid Embase (Elsevier, the Netherlands) and the Cochrane Central Register of Controlled Trials were searched using a combination of appropriate Medical Subject Headings (MeSH) and keywords. The searches were run between August and October 2022. All studies published to date within the medical literature were included, with no limits on date of publication for the majority of searches.

The supplementary material includes the full search strategy for each clinical question. Once the search had been run, further potentially eligible articles were identified by reviewing the reference lists of identified papers. The search was repeated in June 2023 to identify recently published papers and the statement was updated accordingly.

Title, abstract and full texts were screened independently for inclusion by subgroup members and were included based on prespecified eligibility criteria (supplementary material). Any queries or disagreements were resolved through discussion at TF virtual meetings, with final decisions made by the TF chairs (G. Cardillo and N.M. Rahman). Subgroups prepared drafts summarising the relevant literature for their clinical question and, in some cases, algorithms that described the practice of TF members based on

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

extracted data from the evidence presented, which then underwent review by the full TF before being revised and submitted to the chairs.

The TF chairs collated the drafts into a complete statement and the final draft was approved by all members prior to submission to the ERS and hence represents a statement of the entire TF.

Results

An overall diagnostic approach

What tests are used to categorise a pleural effusion as a transudate?

P: Patients with pleural effusions of any cause
I: Pleural fluid (PF) tests and imaging features
C: Not required for literature search (but can use the "Actual Final Diagnosis" as the benchmark against which the investigations are compared against)
O: Diagnostic accuracy, sensitivity, specificity, false-negative and false-positive rates

Summary

- Light's criteria have been shown to be effective at not missing exudates, but owing to moderate specificity (70%), misclassification of transudates as exudates is found to commonly occur (∽25% misclassification rate).
- To overcome this gap, alternative tests have been suggested over the last few decades, but none has been shown to be superior to Light's criteria in detecting exudates correctly.
- When HF is highly suspected but Light's criteria suggest an exudate, a serum-effusion albumin gradient is useful. A result >1.2 $g \cdot dL^{-1}$ indicates the pleural effusion can be accurately reclassified as a transudate due to HF.
- N-terminal pro-brain natriuretic peptide (NT-BNP) serum or PF levels >1500 $\mu g \cdot mL^{-1}$ are accurate at diagnosing HF as the cause of pleural effusion.
- In the presence of a high pre-test probability for liver failure, a PF to serum albumin ratio <0.6 confirms HH when Light's criteria are ambiguous.
- If a serum sample is not available, a PF lactate dehydrogenase (LDH) >67% the upper limit of normal or PF cholesterol >55 $mg \cdot dL^{-1}$ has a high discriminative capacity in diagnosing exudates, equivalent to that of Light's criteria.

In total, 1534 studies were screened to identify 43 studies relevant to producing this section. These consisted of two editorials, one guideline, 27 retrospective observational studies, 11 prospective observational studies and two systematic reviews and meta-analyses (supplementary material).

Why distinguish transudates from exudates?

Even though there are multiple causes of pleural effusion, only a few are responsible for the majority of cases in clinical practice [7, 8]. The most frequent causes of pleural effusion include HF (29%), malignancy (26%), pneumonia (16%), tuberculosis (6%), post-surgery (4%), pericardial diseases (4%) and cirrhosis (3%) [9, 10]. Aetiologies will vary according to whether the effusion is unilateral or bilateral. In a large series by Porcel *et al.* [9], the most common cause of bilateral effusions were HF (53.5%), malignancy (18%) and pericardial disease (7%). The series by Walker *et al.* [4] demonstrated similar findings, with HF and renal failure more likely to present with bilateral effusions (19.8% and 23.1%, respectively) compared to pleural infection (9%).

Given that the causes of pleural effusion are varied and broad, establishing the characteristics of PF is considered the first key step in the diagnostic approach. Classification of pleural effusion as transudate or exudate reflects the pathophysiological mechanisms explaining fluid formation. More than 80% of transudates are due to HF, followed by liver cirrhosis (10%), hypoalbuminaemia, nephrotic syndrome and atelectasis (table 1) [10]. Most transudates can be successfully treated with diuretics, making further investigations unnecessary. By contrast, patients with exudates warrant additional diagnostic procedures to rule out specific important causes (*e.g.* malignancy) or conditions requiring urgent specific treatment (*e.g.* pleural infection) [11].

It should be noted, however, that categorisation of a pleural effusion as transudate or exudate is not always indicative of a particular aetiology or group of aetiologies. Thus, while some pleural effusions are always exudative (*e.g.* tuberculosis) [12, 13], misclassification of cardiac and liver transudates as exudates is relatively common (∽25–30% misclassification rate) [8]. Some conditions may cause either transudate or exudate (*e.g.* non-expansile lung, chylothorax, superior vena cava syndrome) [5]. Finally, other pleural

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

TABLE 1 Causes for pleural effusions according to laterality (unilateral *versus* bilateral) and incidence

	Transudates	Exudates
Common causes	• Heart failure • Liver cirrhosis	• Malignancy • Pneumonia • Tuberculosis • Post-surgery (cardiothoracic, abdominal) • Pericardial diseases
Less common causes	• Hypoalbuminemia • Nephrotic syndrome • Pulmonary arterial hypertension • Atelectasis • Volume overload (*e.g.* in ESRF) • Non-expandible lung[#] • Peritoneal dialysis	• Trauma • Idiopathic • Pulmonary embolism[¶] • Abdominal diseases • Autoimmune diseases • Uremic pleural effusion
Rare causes	• Superior vena cava syndrome[#] • Constrictive pericarditis[#] • Urinothorax[#] • Cerebrospinal fluid • Non-cirrhotic portal hypertension • Extravascular migration of CVC	• Oesophageal perforation • Chylothorax[¶] • Gynaecologic conditions (*e.g.* Meigs syndrome, endometriosis, OHS) • Drugs • Benign asbestos pleural effusion • Viral • Sarcoidosis • Amyloidosis[¶] • Thoracic radiotherapy

Reproduced from Porcel *et al.* [10] with permission. ESRF: end-stage renal failure; OHS: ovarian hyperstimulation syndrome; CVC: central venous catheter. [#]: may also be exudates; [¶]: may also be transudates (when non-Light's criteria have been used).

effusions that are typically exudative may, rarely, fulfil the criteria for a transudate (*e.g.* 3–4% of malignant pleural effusions (MPEs)) [14–16]. The complexity is compounded further when we consider that multiple aetiologies may account for up to 30% of pleural effusions [17]. Therefore, there is no substitute for thorough assessment, integration of clinical features and findings from investigations and, in certain circumstances, trials of therapy before determining aetiology.

Pleural fluid tests for differentiating transudates from exudates

Since the landmark study by Light *et al.* [18], measurement of PF protein and LDH, which are associated with pleural microvascular permeability and inflammation, respectively, has been adopted into routine clinical practice to distinguish between transudates and exudates [19].

The most recent data indicate that Light's criteria (table 2) yield the following operating characteristics for identifying exudates [10]: sensitivity 98%, specificity 72%, positive likelihood ratio (LR) 3.5, and negative LR 0.03. One of the likely reasons explaining the specificity of Light's criteria when applied to real-world clinical practice is that the criteria were originally conceived to maximise the detection of exudates to reduce the risk of failing to diagnose potentially serious conditions (*e.g.* malignancy, infection). To overcome some of the limitations of Light's criteria (*i.e.* low specificity, need of serum samples for ratio calculations, application of dichotomous cut-off values to continuous variables), alternative tests have been suggested over the last few decades, but none has been shown to be superior. If a serum sample is not

TABLE 2 Light's criteria to classify a pleural effusion as an exudate [20]

Parameter	Thresholds[#]
PF to S protein ratio	>0.5
PF to S LDH ratio	>0.6
PF LDH	>0.67 upper limit of normal S value

PF: pleural fluid; S: serum; LDH: lactate dehydrogenase. [#]: meeting any one of the criteria constitutes an exudative effusion.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

available, using an "or" rule of LDH >67% the upper limit of normal serum LDH and cholesterol >55 $mg \cdot dL^{-1}$ in PF has been shown to have a discriminative capacity equivalent to that of Light's criteria [10, 21–23]. A Bayesian approach, considering the pre-test probability of a transudate or exudate, followed by the application of continuous LR for the elements of Light's criteria to calculate post-test probabilities, although elegant, is cumbersome and adds little to the interpretation of test results compared to clinical judgment [24, 25].

When Light's criteria provide results close to the cut-off points for an exudate in the presence of a high pre-test probability for HF or cirrhosis, an albumin gradient (serum albumin minus PF albumin) >1.2 $g \cdot dL^{-1}$ or an albumin ratio (PF albumin divided by serum albumin) <0.6 has been shown to correctly reclassify ∼80% of these "false" exudates [26]. An alternative to identify HF-related effusions is the measurement NT-proBNP in PF (or serum, if thoracocentesis is not planned). Several diagnostic accuracy studies dedicated to assessing the role of natriuretic peptides in the blood (especially NT-proBNP) for diagnosing pleural effusion of cardiac origin have been performed [27–35]. Generally, the studies have demonstrated high sensitivity and specificity. A meta-analysis in 2015 [32] reported a pooled sensitivity and specificity of an elevated blood NT-proBNP for diagnosis of HF as the cause of pleural effusion of 0.92 (95% CI 0.86–0.95) and 0.88 (95% CI 0.77–0.94), and PF NT-proBNP sensitivity and specificity of 0.94 (95% CI 0.90–0.96) and 0.91 (95% CI 0.86–0.95), respectively. The area under the curve on the summary receiver operating characteristic curves for blood NT-proBNP and PF NT-proBNP (accounting for differing thresholds of "normal" range) were 0.94 and 0.96, respectively [32]. Levels >1500 $\mu g \cdot mL^{-1}$ are the most accepted threshold with a positive LR of 7.8 (95% CI 3.7–16.3) and negative LR of 0.10 (95% CI 0.06–0.16) in blood NT-proBNP and a positive LR of 10.9 (95% CI 6.4–18.6) and negative LR of 0.07 (95% CI 0.04–0.12) in PF NT-proBNP (figure 1) [34].

What are the scoring systems for categorising a pleural effusion as a transudate?

P: Patients with pleural effusions of any cause
I: Composite scoring systems/risk prediction tools
C: Not required for literature search (but can use the "Actual Final Diagnosis" as the benchmark against which the investigations are compared against)
O: Diagnostic accuracy, sensitivity, specificity, false-negative rates

Summary

- When faced with pleural effusions categorised as an exudate by Light's criteria, studies have shown a combined clinical-radiological scoring model (table 3) can correctly identify HF as the cause.
- Mean attenuation values on computed tomography (CT) or patterns of echogenicity on thoracic ultrasound (TUS) have been shown to be unreliable for transudate–exudate discrimination.
- Studies have demonstrated that radiological findings alone cannot replace biochemical analysis of PF in combination with clinical data when attempting to categorise a pleural effusion as a transudate or exudate.

Transudate–exudate differentiation: scoring systems and imaging studies

Scarce literature has focused on the creation of a scoring system based on simple and available clinical data which allows distinction between transudate and exudate, or identification of falsely categorised transudates. Porcel *et al.* [36] conducted a study in which they designed and validated a clinical scoring model for identifying pleural effusions due to HF, intended to be applied when effusions meet Light's criteria for exudates (table 3). A score ⩾7 greatly increased the probability of a cardiac aetiology (diagnostic accuracy 92%, LR positive 12.7, LR negative 0.39) and proved to exceed the discriminative capabilities of protein and albumin gradients in isolation [36].

The role of imaging as a noninvasive method for differentiating transudative and exudative pleural effusions has been addressed in several studies.

The most obvious feature that may help differentiate a transudative cause from an exudative cause is the size of the effusion. Massive pleural effusions tend to be seen in malignancy (*i.e.* exudative effusion); however, HHs can also present with sizeable effusions, and therefore this is not an absolute rule and does not replace the need for further assessment [37, 38].

Chest CT is commonly used to evaluate patients with pleural effusion of undetermined aetiology [7, 11]. Under the rationale that exudative effusions have a greater content of protein, LDH and cholesterol, studies

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

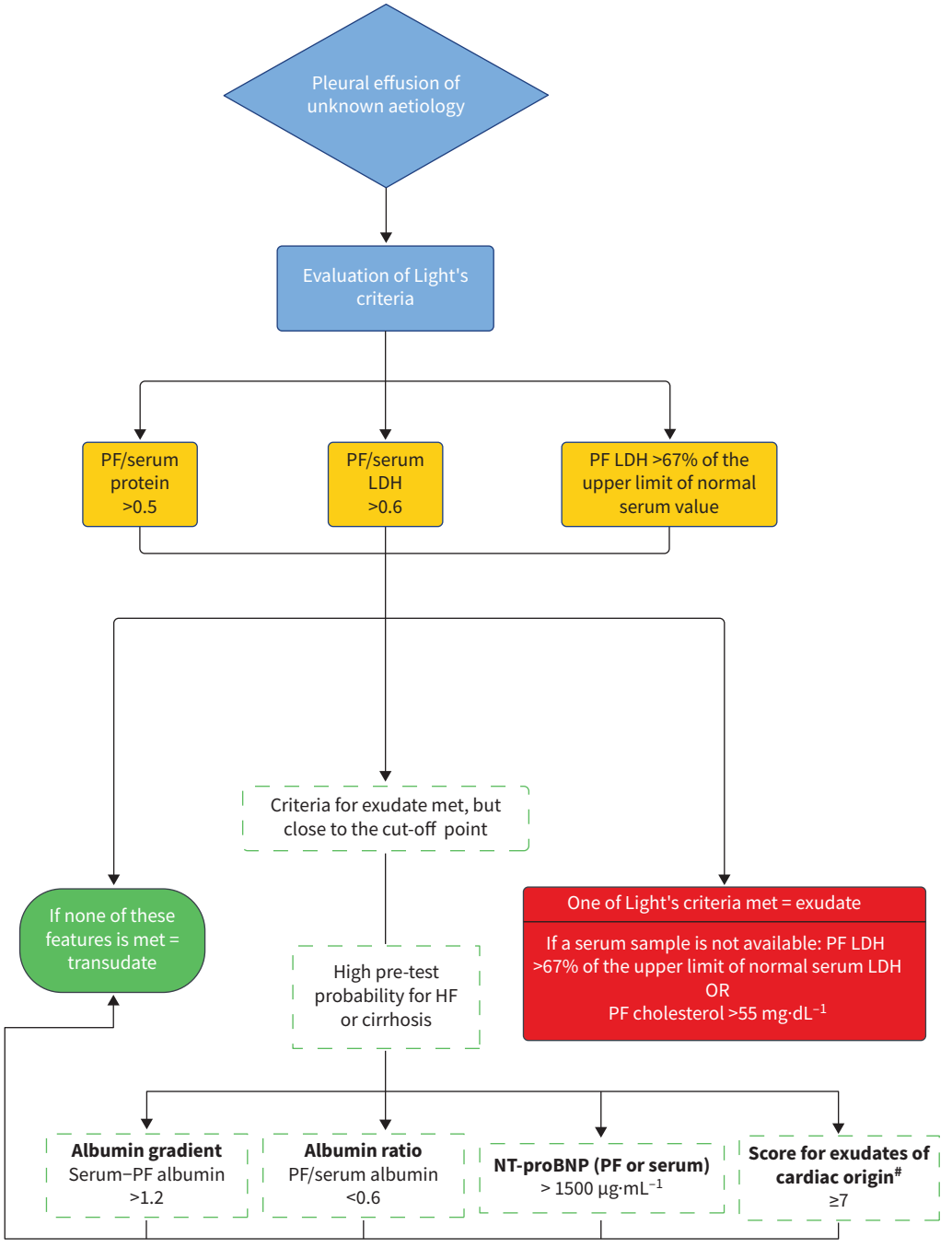


FIGURE 1 A visual depiction of the Task Force members' practice for evaluating a pleural effusion of unknown aetiology. This describes the current practice of the Task Force, and is not intended as a recommendation for clinical practice. PF: pleural fluid; LDH: lactate dehydrogenase; HF: heart failure; NT-proBNP: N-terminal pro-brain natriuretic peptide. #: from Porcel *et al.* [36].

have addressed whether these biochemical characteristics translate to greater radiodensity and higher attenuation values of PF measured in Hounsfield units (HU). Although most studies show that exudates have higher mean attenuation values than transudates, this parameter is not useful for discrimination because of the significant overlap of HU values between transudates and exudates [39–44]. For example, in the largest study (n=127 exudates and 100 transudates based on underlying disease) at the best threshold value of ⩾4 HU, CT identified exudates with a sensitivity of 69%, a specificity of 66%, a LR positive of 2 and a LR negative of 0.47 [44].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

TABLE 3 Scoring system for diagnosing pleural effusions secondary to heart failure in exudates

Parameters	Score
Age ⩾75 years	3
Albumin gradient[#] >1.2 g·dL^{-1}	3
PF LDH[¶] <250 U·L^{-1}	2
Bilateral pleural effusion on chest X-ray	2
Protein gradient[+] >2.5 g·dL^{-1}	1

Reproduced from Porcel *et al.* [36] with permission. PF: pleural fluid; LDH: lactate dehydrogenase. [#]: serum albumin minus PF albumin; [¶]: this figure represents two-thirds of the upper limit of normal for serum LDH in the laboratory the score was derived in; [+]: serum protein minus PF protein.

Chest CT can identify thickening and pleural nodules and provide information about the degree of organisation and loculation of a pleural effusion. However, although these alterations are more frequently seen in exudative effusions, they are not exclusive to exudates and are therefore not reliable parameters to confidently discriminate between transudate and exudate [45, 46].

The echogenicity of PF on TUS has been explored to discern transudate from exudate. Pleural effusions can be categorised as simple (or anechoic) or complex (hyperechoic), and categories within complex include complex non-septated, complex septated and homogenously echogenic. Combining data from five series, comprising 560 transudates and 672 exudates [47–51], demonstrates that an anechoic sonographic pattern has a sensitivity of 80%, specificity of 63%, LR positive of 2.16 and LR negative of 0.32 for transudates. When transudates fall into the complex sonographic category, they are mostly complex non-septated [49, 50]. The overlapping patterns of echogenicity make this feature unreliable for transudate–exudate discrimination. Therefore, using the echogenicity of PF alone to decide whether to aspirate a pleural effusion is questionable based on the literature to date. Conversely, other ultrasound characteristics (*e.g.* pleural nodularity) strongly support the diagnosis of MPE (sensitivity 42.5% and specificity 96.9% in a recent meta-analysis of seven studies) [52]. Figure 2 illustrates some of these features.

Magnetic resonance imaging (MRI) allows morphological and functional imaging of the pleura and can be of value in detecting pleural malignancy and local tumour invasion in mesothelioma cases [53]. However, MRI has no practical applicability in determining the transudative or exudative nature of a pleural effusion. Transudates are more often T2-weighted hyperintense and return low signal intensity on T1-weighted images [54]. Moreover, on diffusion-weighted magnetic resonance imaging (DWI) protocols, the apparent diffusion coefficient is generally slightly lower in exudates than in transudates [55]. In short, no imaging replaces the biochemical analysis of PF in combination with clinical data to categorise a pleural effusion as transudate or exudate.

Areas of future research

Diagnostic accuracy:

- What features suggest an effusion has been falsely labelled as an exudate and what features might help reclassify them?

Diagnostic overlap and integration:

- How can we integrate clinical information, radiography, PF biochemistry and bedside ultrasound to produce scoring systems to better diagnose the aetiology of pleural effusions?

Heart failure

What are the management options for refractory heart failure-related effusions?

P: Patients with refractory HF-related effusions

I: Pleural interventions (therapeutic aspiration/indwelling pleural catheter (IPC)/intercostal drain (ICD)+slurry/poudrage/video-assisted thoracoscopic surgery (VATS)), cardiological interventions (diuresis, fluid restriction, cardiac resynchronisation therapy device, valvular surgery, dialysis)

C: Not required

O: Patient-reported outcome measures (PROMs) on quality of life (QoL), visual analogue scale dyspnoea scores, hospital attendance rates, complications from pleural interventions (*e.g.* bleeding, infection), complications to planned surgery as a result of pleural interventions (*i.e.* due to pleurodesis)

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Summary

- In cases of recurrent HF-related pleural effusions refractory to medical treatment, evidence has demonstrated effective palliation of symptoms with pleural interventions.
- Ultrasound-guided thoracocentesis, IPCs, pleurodesis and, more rarely, surgical procedures and pleuro-peritoneal or pleuro-venous shunting have been described in the literature as pleural procedures for symptomatic relief of recurrent pleural effusions secondary to HF.
- The results of the only randomised controlled trial (RCT) that compared the use of IPCs with repeated thoracocenteses for refractory transudative effusions showed no advantage in dyspnoea relief with the

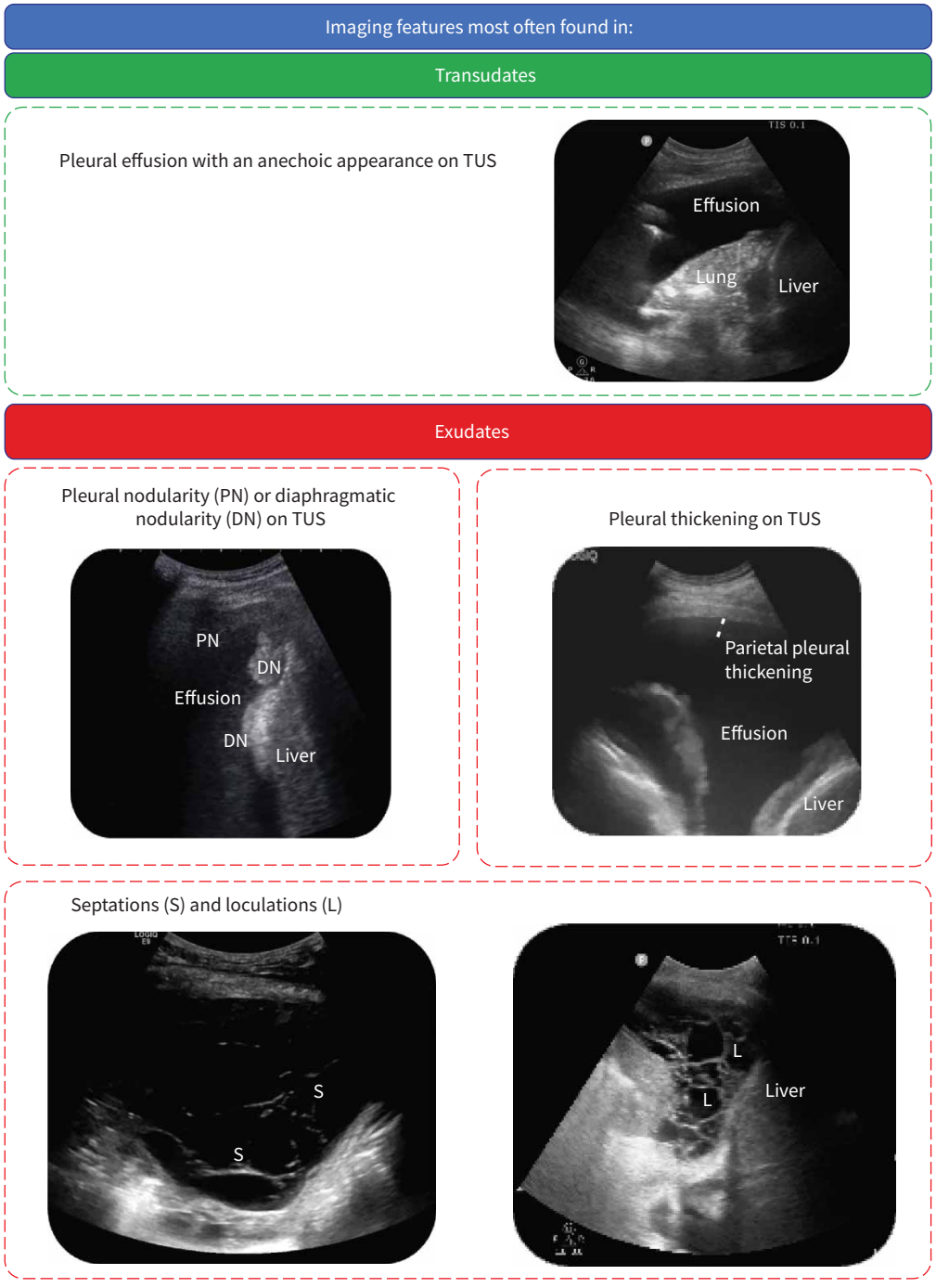


FIGURE 2 Imaging features seen in transudates and exudates (suggestive but not diagnostic). TUS: transthoracic ultrasound.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

use of IPCs and a greater adverse event rate. However, several retrospective studies suggest that IPCs provide palliation of symptoms and reduced length of hospital stay and may be beneficial in patients who require frequent thoracocenteses (three or more events).
- The authors' practice is to perform repeat pleural aspiration for recurrent symptomatic cardiac effusions refractory to medical therapies, with consideration of other treatments only if this strategy does not work or frequent re-intervention is required.
- The literature suggests that IPCs were comparable to talc poudrage for the palliation of patients with recurrent pleural effusions due to HF results in fewer adverse events.

In total, 1034 studies were screened to identify 34 studies relevant to producing this section. These consisted of two editorials or narrative reviews, two guidelines, 12 retrospective observational studies, six prospective observational studies, four non-randomised interventional (comparator) studies, three RCTs, two case series and three systematic reviews and meta-analyses (supplementary material).

Review of the evidence

There are no studies comparing cardiological and pleural interventions for patients who present with recurrent pleural effusions due to HF. However, there is consensus in the literature that pleural interventions should only be considered in patients with symptomatic HF-related pleural effusions refractory to medical treatment [56–69]. The definition of "refractory" is poorly described in the literature. The authors would consider patients with persistent effusions despite maximal tolerated doses of diuresis as "refractory". With the advent of novel pharmacological agents (*e.g.* sodium-glucose co-transporter-2 inhibitors), the incidence of HF-related pleural effusions is expected to decrease; nonetheless, pleural intervention is expected to play an ongoing role in management.

Ultrasound-guided thoracocentesis and chest tube drainage are feasible and low-risk procedures that relieve symptoms in patients with HF [70, 71]. There are no studies investigating complications of these procedures specifically for patients with HF-related pleural effusions. A comprehensive review of general pleural procedure-related complications can be found elsewhere [72].

12 studies reporting the use of IPCs for recurrent pleural effusions secondary to HF were identified (supplementary material: table A) [56–69]. Most studies report symptomatic palliation, reduced hospital admissions and a low to moderate rate of serious adverse events. There were no studies investigating the optimal drainage frequency and volume for NMPEs. In practice, common drainage frequency to begin with is three times per week and around 500–1000 mL PF per session. However, two randomised trials found that daily drainage in patients with MPE drove auto-pleurodesis [73, 74]. The most common IPC complications include pneumothorax, IPC malfunction, IPC-related pain and infections, with empyema and drain site infection being the most prevalent [3, 58, 59, 61, 62, 64, 65, 67]. A systematic review and meta-analysis reported spontaneous pleurodesis in 42% of HF-related pleural effusions [60].

The only RCT to date to evaluate the outcomes of therapeutic thoracocentesis (TT) was the randomised trial of IPCs for refractory transudative pleural effusions (REDUCE) by Walker *et al.* [56]. This was an open label study from 13 British centres which randomised patients with symptomatic pleural effusions secondary to cardiac, renal or hepatic failure to undergo either IPC insertion or sequential TT, where TT was defined as "standard care". No significant difference was reported for the primary outcome of breathlessness for the two procedures, despite greater PF drainage for the IPC group (17.4 L *versus* 2.9 L over 12 weeks). Furthermore, a higher rate of adverse events was observed in the IPC arm (59% *versus* 37%) and IPC use was associated with a higher loss of serum albumin during treatment. Patients who received an IPC underwent fewer additional invasive pleural procedures but they did require lifelong drainage several times per week.

Pleurodesis using external chemical agents has been mostly investigated in the context of MPEs. Nonetheless, different pleurodesis agents have been used for the palliation of recurrent NMPE as well, with talc being the best described. Pleurodesis with ungraded talc with small particles (<10 μM) has been shown to be less safe in a randomised trial owing to significantly greater levels of systemic inflammation and poorer gas exchange [75]. Talc may be delivered as a slurry through a chest drain or *via* thoracoscopy as poudrage. Retrospective studies report a success rate of 75–80% for the use of talc pleurodesis in recurrent NMPE, including in patients with HF [76, 77].

There were two retrospective studies comparing the management of cardiac-related pleural effusions with talc poudrage with or without IPC placement *versus* IPC alone [64, 66]. There was no significant difference in

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

symptom relief between the two techniques. Talc poudrage was associated with longer hospital stay and higher readmission rate, mortality and morbidity. Pleurodesis was achieved at a higher rate in patients who received talc poudrage; thus, patients with IPC alone had a longer time to catheter removal.

Pleuro-peritoneal or pleuro-venous shunting of PF from the pleural space for NMPEs was described in case series [3, 78, 79]. In only two cases, the pleuro-peritoneal devices achieved effective symptomatic relief and required short hospitalisation, while no serious complications were reported [3, 79]. However, this technique is contraindicated in patients with ascites.

Surgical approaches are rarely attempted for patients with HF-related recurrent pleural effusions due to frailty and comorbidity. The authors have on occasion considered surgical pleurectomy in a carefully selected subgroup of patients where noninvasive techniques have failed or are contraindicated, especially where there is trapped lung [3].

Areas of future research

Treatment options:

- What is the optimal pleural intervention for patients with HF-related pleural effusions: IPC (±talc slurry pleurodesis), repeat thoracocentesis, talc pleurodesis?

Optimising existing treatment paradigms:

- What is the optimal drainage frequency and volume for patients with HF-related pleural effusions and an IPC *in situ*?
- What is the safe maximum drainage volume during thoracocentesis in a patient with HF-related pleural effusion?

What investigations are used for a unilateral effusion in a patient with known heart failure?

P: Patients with known (decompensated) HF and presenting with a unilateral effusion

I: Stratification to conservative management or invasive diagnostics according to clinical algorithms (using noninvasive tests: serum brain natriuretic peptide, TUS)

C: Not required

O: Complications arising from invasive investigations (bleeding, infection, pneumothorax), outcomes (mortality, risk of circulatory or respiratory failure), delayed diagnosis of clinically relevant differential diagnosis (*e.g.* MPE, pleural infection)

Summary

- Several studies have assessed the diagnostic accuracy of noninvasive tests such as serum natriuretic peptides or bedside ultrasound in patients with unilateral pleural effusion of unknown origin.
- However, no study to date has addressed a patient population with known HF and unilateral pleural effusion stratified or randomised to conservative management or invasive diagnostics based on clinical algorithms using serum natriuretic peptides or bedside ultrasound.
- Several studies have demonstrated acceptable diagnostic accuracy for identifying patients with a cardiac pleural effusion in unselected patients with unilateral pleural effusion.
- The authors adopt a pragmatic approach using a combination of serum tests and ultrasound to guide management in patients with unilateral pleural effusion and known HF to ensure a reasonable balance between the risk of missing a non-cardiac cause of pleural effusion and the risks of performing an invasive diagnostic procedure (*i.e.* thoracocentesis) (figure 3 and table 4).

Review of the evidence

Patients with known HF presenting with a unilateral pleural effusion is a clinical dilemma for the treating physician and the patient. While not considered typical for HF, in one large series of patient presenting with acute decompensated HF and pleural effusion (n=1504), pleural effusions were unilateral in 41% of cases [80]. Potential benefits of performing additional diagnostic and invasive procedures are 1) a more accurate determination of whether the pleural effusion is caused by decompensated HF and, more importantly, 2) minimising the risk of a delayed clinically relevant differential diagnosis such as MPE and bacterial pleural infection. There is often also a belief that a hypoxaemic patient with a pleural effusion, albeit transudative, may benefit from drainage. However, there is now consistent evidence to the contrary that a pleural effusion tends not to cause hypoxaemia and drainage rarely leads to correction, outside of specific settings (*e.g.* large bilateral effusions) [81].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

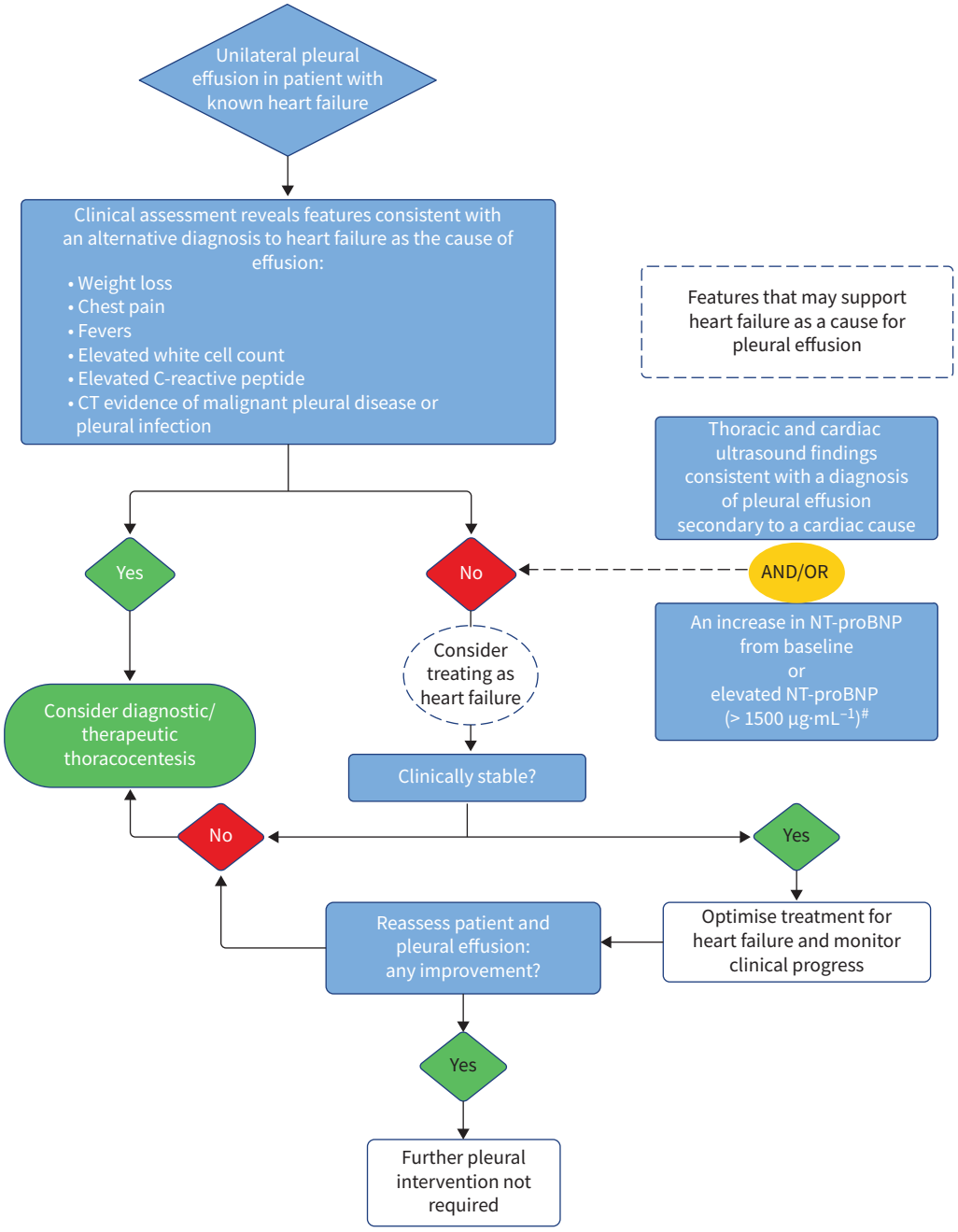


FIGURE 3 A visual depiction of the Task Force members' practice when assessing a unilateral pleural effusion in a patient with known heart failure. This describes the current practice of the Task Force, and is not intended as a recommendation for clinical practice. CT: computed tomography; NT-proBNP: N-terminal pro-brain natriuretic peptide. #: the cut-off value of 1500 µg·mL^{-1} was derived from observational studies differentiating heart failure as the cause of effusion compared to alternative diagnoses, rather than unilateral effusions presenting in patients with known heart failure.

The potential drawback of additional diagnostic and invasive procedures is the increased risk of complications from invasive procedures (*e.g.* bleeding, infection, pneumothorax), increased costs and time use, and a delay in goal-directed therapy in known cardiac disease.

Of special interest is whether an approach based on noninvasive tests can be used to accurately identify a subgroup of patients in which diagnostic and therapeutic invasive procedures (*e.g.* thoracocentesis) can safely be avoided. The multi-organ impairment often seen in patients with HF and the concomitant

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

TABLE 4 Noninvasive bedside test findings in patients with heart failure and unilateral pleural effusion

Modality	Findings supporting pleural effusion of cardiac origin	Findings suggesting pleural effusion of non-cardiac origin
Echocardiography		
Cardiac findings	• Findings consistent with systolic or diastolic heart failure • Other findings consistent with pleural effusion of cardiac origin (*e.g.* severe valve abnormality)	• No apparent findings consistent with heart failure or other cause for cardiac decompensation • Findings suggesting malignancy, inflammation or infection (*e.g.* large pericardial effusion, complex pericardial effusion, visible tumour) • Findings suggesting possible thromboembolic disease (*e.g.* visible thrombus, D-sign, McConnell's sign)
Assessment of inferior vena cava	• Signs indicating grossly elevated central venous pressure	• No obvious signs of elevated central venous pressure
Thoracic ultrasound	• Presence of interstitial syndrome • Simple pleural effusion • Sono-morphology consistent with simple compression atelectasis (in case of visible lung consolidation)	• Absence of interstitial syndrome • Complex pleural effusion • Findings suggesting possible malignant pleural effusion (*e.g.* parietal pleural or diaphragmatic thickening/nodularity, lung parenchymal pathology not typical for simple compression atelectasis) • Findings suggesting possible thromboembolic disease (*e.g.* hypoechoic pleural-based lesion)
Serum NT-proBNP	• $\geqslant$1500 μg·mL^{-1}	• <1500 μg·mL^{-1}

NT-proBNP: N-terminal pro-brain natriuretic peptide.

comorbidities and overlapping risk factors (*e.g.* smoking) make it a clinical challenge to isolate a single most likely aetiology for unilateral pleural effusion.

Several blood biomarkers have been assessed to diagnose HF and pleural effusion of cardiac origin (see "An overall diagnostic approach"). Measurement of natriuretic peptides in the blood such as B-type natriuretic peptide (BNP) or NT-proBNP are considered standard in the initial assessment of patients with suspected HF, especially as a rule-out tool [82]. Different cut-off points to define a positive test result have, however, been used and some studies were performed in selected patient populations (*e.g.* patients with pleural transudates). The systematic literature search did not reveal any studies in which the role of natriuretic peptides had been specifically assessed in a population comprising unselected patients with known HF and in which the cause of a unilateral pleural effusion had to be determined. Based on the currently published literature, natriuretic peptides are most likely of clinical utility in patients with known HF and a unilateral pleural effusion. Low values usually raise suspicion of an alternative cause of pleural effusion, whereas high values support a cardiac origin, provided features of an alternative diagnosis are not present (figure 3).

Imaging

As a noninvasive technique for determining causes of pleural effusion, several imaging modalities have been studied in the context of patients with possible HF [83, 84]. The most commonly studied modality has been ultrasound, with the systematic search identifying a single study assessing the role of more advanced imaging [85].

Echocardiography remains the standard initial diagnostic imaging modality in patients with suspected HF [82]. For patients with HF, echocardiographic-assessed parameters such as systolic pulmonary artery pressure and E/A ratio (early to late ventricular filling ratio) have been shown to be risk factors for the formation of pleural effusion [86]. Focused cardiac ultrasound performed by non-cardiologists has been shown to have an acceptable diagnostic accuracy for diagnosing systolic left-sided HF [87]. Sonographic features suggestive of decompensated HF with cardiogenic pulmonary oedema are presented in table 4 [88–94]. While interstitial syndrome *per se* is not specific for cardiogenic pulmonary oedema, when integrated with the clinical assessment and other findings it outperformed NT-proBNP, chest X-ray and clinical assessment alone in differentiating acute decompensated HF from non-HF-related causes for dyspnoea. The additional benefit of ultrasound is the ability to detect alternative causes for pleural effusion (*e.g.* MPE, pulmonary embolism, pneumonia) in the presence of specific findings [95–98].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Despite the many studies assessing the role of bedside ultrasound for diagnosing HF, decompensated HF with cardiogenic pulmonary oedema and non-cardiogenic causes of pleural effusions, the systematic literature review did not identify any studies in which bedside ultrasound had been used in a setting of patients with known HF and with the diagnostic dilemma of a unilateral pleural effusion.

Although studies assessing the use of ultrasound to determine the cause of unilateral pleural effusion in patients with HF are lacking, studies assessing the use of TUS to monitor patients with HF have been demonstrated to have a potential clinical impact on clinically relevant outcomes [93, 99, 100]. In two RCTs, patients with HF were randomised to therapeutic strategies according to a usual care approach *versus* a lung ultrasound-guided approach. The primary end-points assessed cardiac adverse events and comprised cardiovascular death, readmission or emergency department or day hospital visit due to worsening HF at 6 months. The studies differed in their findings, with one showing no differences between the interventions and one favouring a lung ultrasound-guided approach [99, 100]. Despite these differences, the study findings may show promise for further trials assessing noninvasive ultrasound-guided diagnostic and therapeutic strategies *versus* usual standard of care.

In summary, currently published literature shows that bedside clinical ultrasound is able to detect features of decompensation in patients with known HF (*e.g.* interstitial syndrome) while demonstrating an absence of features to suggest an alternative cause for a unilateral effusion (*e.g.* parietal or diaphragmatic nodularity), which sometimes aids clinician decision-making on diagnostic and therapeutic approaches (figure 2). Where this does not apply, additional diagnostic tests including invasive procedures are sometimes necessary.

Areas of future research

Diagnostic accuracy:

- What is the diagnostic accuracy of noninvasive bedside testing (*e.g.* ultrasound, NT-proBNP) when used in a selected population of patients with HF and unilateral pleural effusion?

Diagnostic overlap and integration:

- What is the optimal integrated use of multiple noninvasive bedside testing (*e.g.* ultrasound, NT-proBNP) when used in a selected population of patients with HF and unilateral pleural effusion?

Clinical impact:

- Can use of noninvasive bedside testing (*e.g.* ultrasound, NT-proBNP) when used in a selected population of patients with HF and unilateral pleural effusion have a positive impact on clinically relevant outcomes (*e.g.* adverse events, length of hospitalisation, PROMs)?

Hepatic hydrothorax

What are the therapeutic options in patients with symptomatic refractory hepatic hydrothorax?

P: Patients with decompensated chronic liver disease who are eligible for liver transplant and have refractory HH

I: Pleural interventions (therapeutic aspiration/IPC/ICD+slurry/poudrage/VATS), hepatological interventions (transjugular intrahepatic portosystemic shunt (TIPS), albumin infusion, abdominal paracentesis, diuresis)

C: Not required

O: Dyspnoea, QoL, need for re-intervention, hospital length of stay, survival, complications

Summary

- The management of refractory HH can be challenging and multidisciplinary team involvement comprising hepatology, pleural and transplant services is key.
- Hepatic transplantation is considered to be the only curative treatment for HH; the authors' own practice is to refer all eligible patients for liver transplantation.
- TT is effective at providing temporary control of dyspnoea and has a low rate of procedural complications.
- TIPS improves HH in ~50% of cases, although mortality in these patients remains high.
- Both TT and TIPS have been used by the authors to control effusions while awaiting transplantation.
- Data to support chest tube pleurodesis and surgical approaches are limited to small case series with evidence of high complication rates and associated mortality.
- IPCs offer similar control of dyspnoea when compared to repeat thoracocentesis but with a greater rate of complications; if a prolonged transplant waiting list time is expected, provided the risk is acceptable to the patient and clinician, an IPC may be offered.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

- In non-transplant candidates, for those requiring more than three thoracenteses or where repeated thoracocentesis is considered high risk, in the authors' practice an IPC is an acceptable palliative option.

In total, 3260 studies were screened to identify 27 studies relevant to producing this section. These consisted of three editorials or narrative reviews, 19 retrospective observational studies, one prospective observational study, one non-randomised interventional (comparator) studies, one RCT and two case series (supplementary material).

Definition of refractory hepatic hydrothorax

HH is defined as a pleural effusion that occurs in the context of advanced liver disease without an underlying cardiorespiratory cause. It is common, occurring in up to 15% of patients with cirrhosis and portal hypertension and is associated with disabling dyspnoea and poor prognosis [4, 101].

Most patients with HH can be managed with medical therapies alone, which focus on salt restriction and diuretic therapy. However, ~25% are considered to have refractory disease: disease that does not respond to medical management [101, 102]. This frequently requires procedural intervention.

Therapeutic thoracocentesis in hepatic hydrothorax

TT is widely used to provide temporising control of breathlessness in refractory HH and has even been considered "standard care" in research settings [56]. Despite this, there is little high-quality data to guide its use. In the aforementioned REDUCE trial, only 16 out of 68 patients (24%) had HH [56]. Nonetheless, no difference in breathlessness was seen in subgroup analysis of those with HH.

Six non-randomised studies that evaluated TT were identified, involving >1500 patients [103–108]. There were significant variations in study design but the majority were retrospective and no definition of refractory HH was provided. Three did not compare TT against another intervention but used controls who did not have HH to assess outcomes.

Mortality ranged from 18.6% at 30 days [103] to 30% at 180 days [104]. It is likely that these reflect the prognosis associated with HH rather than TT-associated complications because procedure-related complication rates were low (4.9–7.7%) [105, 106].

The largest study evaluating TT for HH was conducted by Hung *et al.* [103], who examined Taiwanese insurance registry data to evaluate survival outcomes in patients who had undergone thoracocentesis for HH, compared to those who underwent "catheter drainage". No precise definition of catheter drainage was given but the authors note that most of these patients underwent "pigtail drainage". There were 1278 patients in each group with propensity matching used to control for known confounders. The 30-day mortality was significantly higher in the group treated with catheter drainage compared to those who underwent thoracocentesis (23.5% *versus* 18.6%, p=0.001). However, the study lacked granular detail, particularly regarding pre-intervention decision-making and it is possible that patients who underwent catheter insertion had more severe or refractory disease, independent of known confounders.

The role of transjugular intrahepatic portosystemic shunt in patients with hepatic hydrothorax

Six retrospective studies evaluated the role of TIPS in the management of HH (supplementary material: table B) [109–114].

Notably, a definition of "refractory" was lacking in five of the six studies and there were significant differences in study design, timing and indication for TIPS. The outcomes included in the analyses were also heterogenous and included the need for post-TIPS thoracocentesis, risk of hospital readmission and TIPS-related adverse events. Some of these outcome metrics were subjective or poorly defined (*e.g.* generic "clinical improvement"). In total, 304 patients underwent TIPS; these patients were mostly male (~60%), with alcohol and/or viral-related cirrhosis (~50%); the median Model for End-stage Liver Disease (MELD) score was 17 points; Child-Pugh stage was not available in most series. Duration of post-TIPS follow-up for outcome assessment ranged between 1 and 6 months. Rates of complete resolution of HH at 6 months were reported in three studies and ranged between 20% and 55% [112–114]. Importantly, only Jindal *et al.* [113] evaluated the association between porto-systemic pressure gradient reduction and resolution of HH, finding no optimal threshold for HH reduction. Three studies evaluated the risk of TIPS-related adverse events; the most common complication was hepatic encephalopathy in 15% of TIPS recipients [112, 113] and patients older than 70 years old were at higher risk of early re-hospitalisation due to hepatic encephalopathy [109].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Notably, only the study by Young *et al.* [114] included patients who received expanded polytetrafluoroethylene (ePTFE)-covered stents; most of these studies included patients who received uncovered TIPS with no data on TIPS diameter. Uncovered stents are associated with a higher risk of stent thrombosis and/or dysfunction; therefore, results from studies of uncovered stents should not be extrapolated to the currently used ePTFE-covered stents, which are likely to achieve better results than uncovered stents. Further studies assessing the efficacy and safety of covered TIPS in patients with HH are expected. Additionally, no clear information on patient selection, *e.g.* exclusion of patients with diastolic dysfunction and/or chronic kidney disease, is available owing to the retrospective design [115].

Mortality after TIPS was between 40% and 50% at 6–12 months [112, 113] and was predicted by severity of liver dysfunction (MELD score), indicating that these patients should be promptly evaluated for liver transplantation, which remains the only curative option for HH.

Current evidence suggests that TIPS may lead to resolution (or at least improvement) of HH in up to 50% of patients within 6 months. Further prospective studies evaluating the use of new ePTFE-covered, controlled-expansion TIPS in the management of HH are eagerly expected.

Chest tube pleurodesis

Two trials evaluating chest tube pleurodesis were identified, including 51 patients in total. Both studies were prospective but there were differences in the definition of refractory HH, sclerosant used, delivery method and outcome measures.

In the largest of these, Mahmoud *et al.* [116] randomised 40 patients following a second presentation of HH one to one to either tigecycline or bleomycin *via* a pigtail catheter. The primary outcome was pleurodesis failure, assessed radiologically at 3 months. This was significantly higher in the tigecycline group with 10 out of 20 (50%) compared to five out of 20 (25%) in the bleomycin group (p=0.031). Tigecycline was also shown to have fewer side effects.

The other study by Lee *et al.* [117] prospectively assessed 11 patients who had not responded to medical treatment. The majority of patients (nine out of 11) underwent chest tube pleurodesis with two patients undergoing VATS pleurodesis. Patients underwent a median of three pleurodesis "sessions". Viscum album was used in six, talc in three and taurolidine in two. Successful pleurodesis, defined as the absence of dyspnoea with no radiological evidence of effusion, was seen in eight out of 11 patients (72.2%) at 1 month although three of these individuals had recurrence within 1 year. Concerningly, five patients (45.5%) were suspected to have procedure-related mortality secondary to acute renal failure.

The inferences that can be drawn from these trials are limited by their scale, single-centre nature, lack of blinding and only a radiological outcome measure.

Indwelling tunnelled pleural catheters

Nine studies of IPCs as the intervention of choice for HH management met inclusion criteria (supplementary material: table C) [56, 59, 61, 65, 69, 118–121]. With the exception of one RCT by Walker *et al.* [56] evaluating the role of IPC *versus* TT and one prospective study of 24 participants [118], all other studies were retrospective. A cumulative total of 269 patients with HH were included in these studies. The definition of refractory HH was lacking in some of the studies and most studies did not have a comparator arm.

Walker *et al.* [56] included 16 patients (eight in each arm) with HH. Although the study sample size of the HH subgroup was small, this is the first study to examine a patient-centric primary outcome in comparison to another intervention in the HH population.

The largest three studies were remarkable for higher risk of complications and lower pleurodesis rates in the HH population compared to historical non-HH effusion data, although no head-to-head comparisons are available to date. Catheter dwell time was not reported in all studies but median time to spontaneous pleurodesis ranged between 55 and 222 days and pleurodesis success was noted in 11–51% [118–120].

Detailed data regarding the IPC placement indication were lacking in most studies. The two largest studies with sample sizes of 62 and 79 patients reported that nine out of nine IPC removals and 11 out of 22 IPC removals occurred after transplant, respectively [119, 120]. Patients who undergo liver transplant with their IPCs still in place may experience resolution of their portal hypertension followed by cessation of excess PF accumulation. In such a scenario, patients may not have undergone pleurodesis *per se*, but simply

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

returned to physiological volumes of PF production, hence meeting the pleurodesis definition criteria used in these studies. This, as well as selection bias, retrospective design and lack of long-term follow-up data, may explain the overestimated pleurodesis rate within the selected studies.

The most common complication in all studies was infection (5–16%), primarily of the pleural space and often in conjunction with catheter site cellulitis. This is slightly higher than that noted amongst the general IPC population (5.8%, 95% CI 5.1–6.7%) [72]. Other complications such as fluid leakage and catheter dislodgement were reported less frequently. A mortality rate of 2.5–5% was reported in the two largest cohorts. One study reported a case of catheter-related sepsis and death in a patient with an IPC as a bridge to transplant [120].

Another study reported a small but statistically significant decrease in body mass index (1.13 $kg \cdot m^{-2}$, p=0.008) and serum albumin levels (0.3 $g \cdot dL^{-1}$, p=0.005) following IPC placement [119]. The optimal drainage volume and frequency, and the role of albumin replacement in PF drainage of HH, are yet to be determined; however, most studies used a symptom-guided or alternate day drainage strategy.

In view of the reported significant adverse event rates, including empyema and death from sepsis following IPC placement in patients with HH, the authors' practice is to avoid this intervention in patients who are considered eligible for a transplant, especially if they are likely to receive a transplant within 3 months. The role of IPC in reducing breathlessness during the first 3 months of IPC placement may be similar to TT but its longer-term benefit in alleviating patient symptoms in unknown. It is the TF members' practice to individualise IPC use in HH following careful evaluation in a multidisciplinary fashion, which may well include palliative care teams while also accounting for patient preference. Figure 4 outlines an approach practised by TF members in managing patients with HH.

Surgical management

Thoracoscopy and pleurodesis, diaphragmatic defect repair

Four retrospective series examining surgical management were included, involving a total of 69 patients. Only one out of four did not have a definition for refractory HH [122]. A variety of sclerosants were used, of which talc was most common, and three out of four trials did not have a comparator arm. Reported success rates varied from 47.6% to 66.6% [123, 124], although there was inconsistency in follow-up periods and efficacy was assessed on the basis of chest X-ray findings rather than PROMs.

Two studies, which involved seven patients in total, also combined diaphragmatic repair with chemical pleurodesis [123, 124]. Milanez de Campos *et al.* [124] reported a 60% success rate in three out of five patients who underwent diaphragmatic closure compared to 43.7% in those who did not. However, it should be noted that two out of five who did not report success had major complications including empyema and death at day 18 post-procedure.

Liver transplant

In the one retrospective single-centre study specifically evaluating the outcome of 28 patients with HH undergoing transplantation (12 out of 18 with refractory HH), post-transplant survival was excellent and comparable to that of matched controls transplanted for other indications. Secondary outcomes, *e.g.* length of surgery, intensive care unit stay, days of mechanical ventilation and transfusion requirements, were also comparable [125].

Areas of future research

Diagnostic accuracy:

- What are the features that predict an alternative diagnosis in patients with presumed HH?

Prognostication:

- What are the clinical and radiographic predictors of complications (specifically including IPC-related infections) in patients undergoing pleural intervention for HH?

Risk prediction:

- Define bleeding risk in the HH population, identify risk factors for bleeding, are there subpopulations which require evaluation of coagulopathy and platelet function/quantity?

Treatments:

- What is the efficacy of talc pleurodesis for HH in prospective comparative trials?
- What is the optimal drainage regimen for patients with HH and an IPC *in situ*?

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

- What is the role of ambulatory aggressive drainage and talc pleurodesis in HH symptom palliation and mortality?
- How should spontaneous bacterial empyema be managed?
- How should ascites be managed in the context of HH?
- What are the relative benefits of TIPS revision *versus* IPC in the subpopulation of patients with delayed TIPS failure?
- What is the efficacy of contemporary covered, small-diameter TIPS?
- What are the indications for TIPS revision?

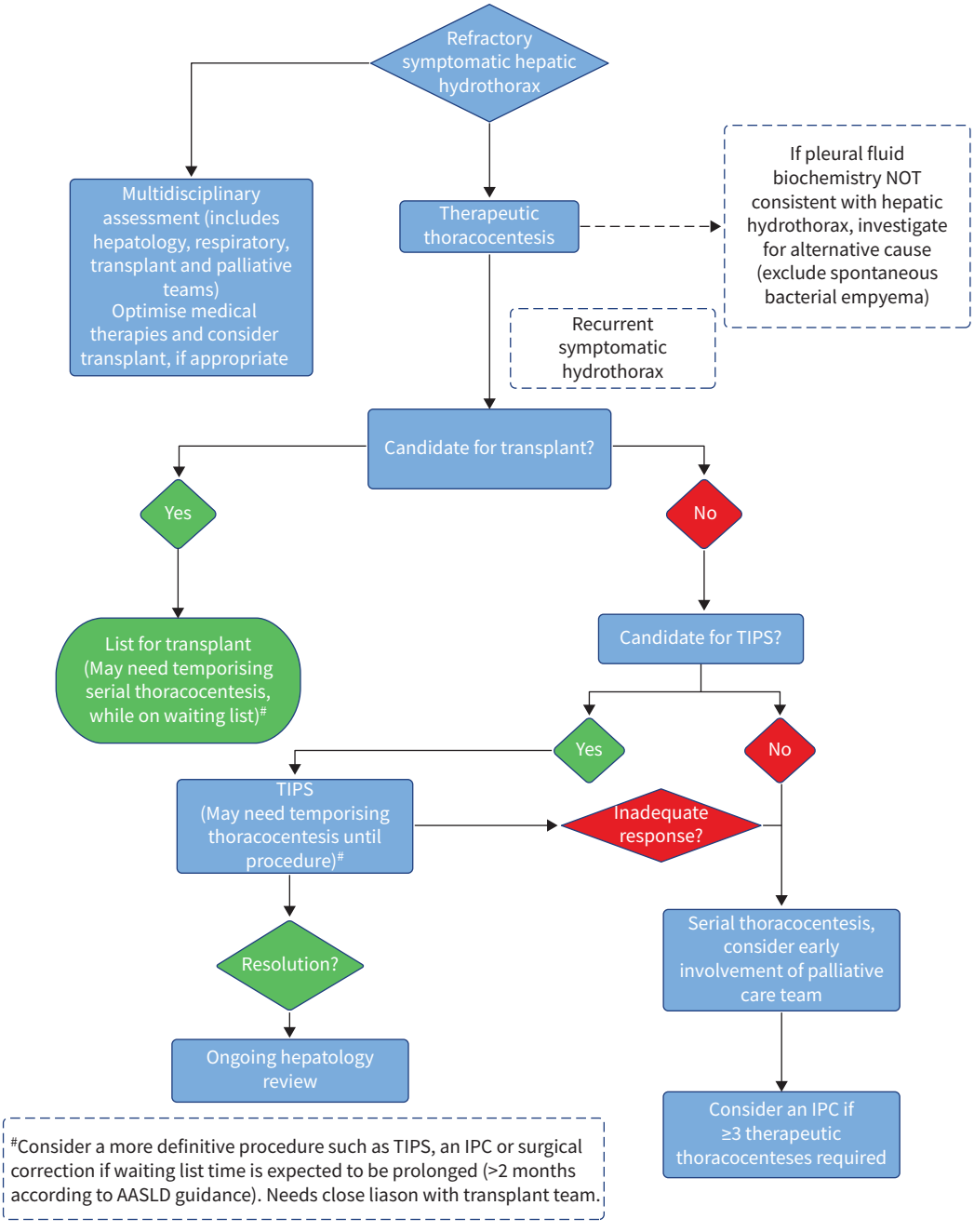


FIGURE 4 A visual depiction of the Task Force members' practice in managing a refractory symptomatic hepatic hydrothorax. This describes the current practice of the Task Force, and is not intended as a recommendation for clinical practice. TIPS: transjugular intrahepatic portosystemic shunt; IPC: indwelling pleural catheter; AASLD: American Association for the Study of Liver Diseases.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

End-stage renal failure

Within the dialysis population, what are the management strategies for effusion control in the event of recurrent effusions?

P: ESRF receiving dialysis

I: Pleural procedures (therapeutic aspiration/IPC/ICD+talc/medical thoracoscopy (MT)/VATS), renal interventions (high-dose diuretics, aggressive fluid removal at dialysis, salt+fluid restriction, use of hypertonic exchanges, use of icodextrin fluid on peritoneal dialysis (PD), switch from PD to haemodialysis)

C: Not required

O: Patient-reported symptom measures, QoL, frequency and duration of dialysis sessions, volume of fluid removed during dialysis (+complications), pleural procedural complications, total number of pleural interventions required, number of breakthrough pleural interventions required

Summary

- Evidence on treating pleural effusions in ESRF is limited, with a single RCT representing only six patients from this population.
- This population is frail, has a poor prognosis and carries a high symptom burden; thus, usually proposed treatments are for palliative intent.
- The commonest aetiology for effusions described in the literature in this population is fluid overload, but not all patients will present with bilateral effusions or even transudates.
- In this population, there is a significant risk of pleural infection or malignancy. Where there is clinical suspicion and further investigation is deemed appropriate, it is the authors' practice to conduct cross-sectional imaging early in the diagnostic pathway.
- If the aetiology is fluid overload, aggressive medical management or renal replacement therapy (RRT) adequately treat pleural effusions. However, in many cases the adverse event rates of aggressive RRT can limit this approach. In contrast, pleural interventions have been shown to be relatively safe across several observational studies.
- The choice of pleural intervention is guided by patient choice and available treatment methods, but similar symptomatic relief has been achieved by repeat thoracocentesis alone when compared with IPCs in observational studies.
- Given the high adverse event rate and increased drainage volume with IPCs seen in RCTs of benign pleural effusions, the authors' practice is to offer serial thoracocentesis as the first treatment option, with IPCs or attempted talc pleurodesis reserved for refractory cases.

In total, 1763 studies were screened to identify 21 studies relevant to producing this section. These consisted of 16 retrospective observational studies, three prospective observational studies and two case series (supplementary material).

Pleural effusions in end-stage renal failure

Pleural effusions in the ESRF population are common and secondary to varying pathophysiological processes. In the series by Jarrat and Sahn [126], the overall incidence of pleural effusion among a hospitalised population receiving haemodialysis was 21%, but only a fraction were attributed to uraemic pleuritis (16%), with the majority ascribed to cardiac impairment (46%). It is noteworthy that this series predated the modern use of echocardiography and the authors did not appear to differentiate HF from hypervolaemia. Recent studies have tended to differentiate between HF and fluid overload and suggest the latter is the leading cause of pleural effusions among hospitalised patients: 9.6% *versus* 61.5%, respectively [127]. A summary of the prevalence of pleural effusions and aetiologies described in the medical literature is presented in supplementary material: table D. Estimates on the prevalence of pleural effusions among patients with ESRF vary (6.7–51%) and are heavily influenced by imaging modality, degree of renal impairment and form of RRT. With this in mind, the estimated prevalence of pleural effusions among the studies reviewed is 1190 out of 4826 (24.7%, 95% CI 23–26%) [126–137].

Patients with ESRF may present with effusions due to many causes:

- hydrostatic and oncotic imbalances
- HF
- fluid overload
- hypoalbuminaemia (secondary to *e.g.* nephrotic syndrome)
- inflammation of the parietal pleura (secondary to *e.g.* autoimmune disease)
- infection or malignancy as a result of immunosuppression (secondary to *e.g.* ESRF or following transplant).

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

TABLE 5 Unusual causes of pleural effusion in end-stage renal failure, with proposed management options from the medical literature

Cause of pleural effusion	Proposed mechanisms	Diagnostic features	Management
Uraemic pleuritis [138, 139]	Mechanism unknown, suggested that toxins produced as a result of uraemia lead to effusion formation or this represents a fibrotic process following bleeding into pleural cavity as a result of circulating heparin (administered as part of RRT) or due to coagulopathy due to uraemia	Usually a diagnosis of exclusion PF is exudative, often haemorrhagic Chronic fibrinous pleuritis in pleural biopsy histology	Increase intensity of RRT Tube thoracostomy ±pleurodesis Pleural decortication Consider systemic corticosteroids
Urinothorax [140]	Urine diverted into pleural cavity through diaphragmatic defects or lymphatic channels as a result of obstructive uropathy or trauma of urinary system	PF may be transudative or exudative (if high LDL), and of low pH, PF creatinine/serum creatinine >1 Renal scintigraphy with ^{99m}Tc ethylene dicysteine (detection of extravasation of the tracer dye and its collection in the pleural cavity)	Surgical or radiological intervention for anatomical defect
Nephrotic syndrome [140]	Low oncotic pressure (due to proteinuria) and increased hydrostatic pressure (due to salt retention) result in increased PF production Hypercoagulability leading to pulmonary embolism may also lead to pleural effusion, as may pleural infection secondary to immunoparesis due to loss of immunoglobulins	PF usually transudative but may be exudative in view of the alternative mechanisms highlighted	Treatment of fluid overload and hypoproteinaemia and direct management of nephrotic syndrome
Vascular abnormalities secondary to complications from haemodialysis [141]	Increase in hydrostatic pressure due to vascular obstruction leads to increased PF formation and decreased lymphatic clearance	Often a unilateral transudative pleural effusion Venogram diagnostic	Ligation of fistula or venoplasty
PD-associated pleuro-peritoneal leak [142] (see following PICO)	Increase intra-abdominal pressures following PD and porosities in the diaphragm lead to pleural effusion formation	Often an extreme transudate with very low protein values (<1 g·dL^{-1}) and very elevated glucose values (PF glucose/serum glucose ratio >1 CT peritoneography or scintigraphy Trial of PD cessation	Alternative mode of RRT Pleurodesis Surgical repair

RRT: renal replacement therapy; PF: pleural fluid; LDL: low-density lipoprotein; PD: peritoneal dialysis; CT: computed tomography; PICO: Population, Intervention, Comparator and Outcomes.

More unusual causes are recognised and are presented in table 5.

Diagnostic implications for pleural effusions in end-stage renal failure

While the usual pathways for investigating pleural effusions apply [19], there are specific areas in regards to the ESRF population that need considering in the diagnostic approach (figure 5).

Management strategies for end-stage renal failure patients who present with effusions

Many of the studies in this review included patients with a myriad of different aetiologies driving pleural effusion, and therefore caution should be exercised in applying the evidence to all patients. It may not always be apparent that an effusion is secondary to fluid overload given some of the dual mechanisms involved, and therefore a pragmatic approach is often applied, with trials of therapies.

However, it is clear that patients with ESRF who manifest a pleural effusion tend to have a poorer prognosis, and a greater degree of cardiac comorbidity and death from cardiovascular disease, when compared to those without pleural effusion [131, 145]. This is supported by observational data from Walker *et al.* [4], who found a 6-month and 1-year mortality of 31% and 46%, respectively, three times that observed in the general ESRF population (1-year mortality 15.6%), as well as from other studies [146, 147]. In addition to poor outcomes, there is a significant symptom burden, with many patients reporting dyspnoea [133, 135].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

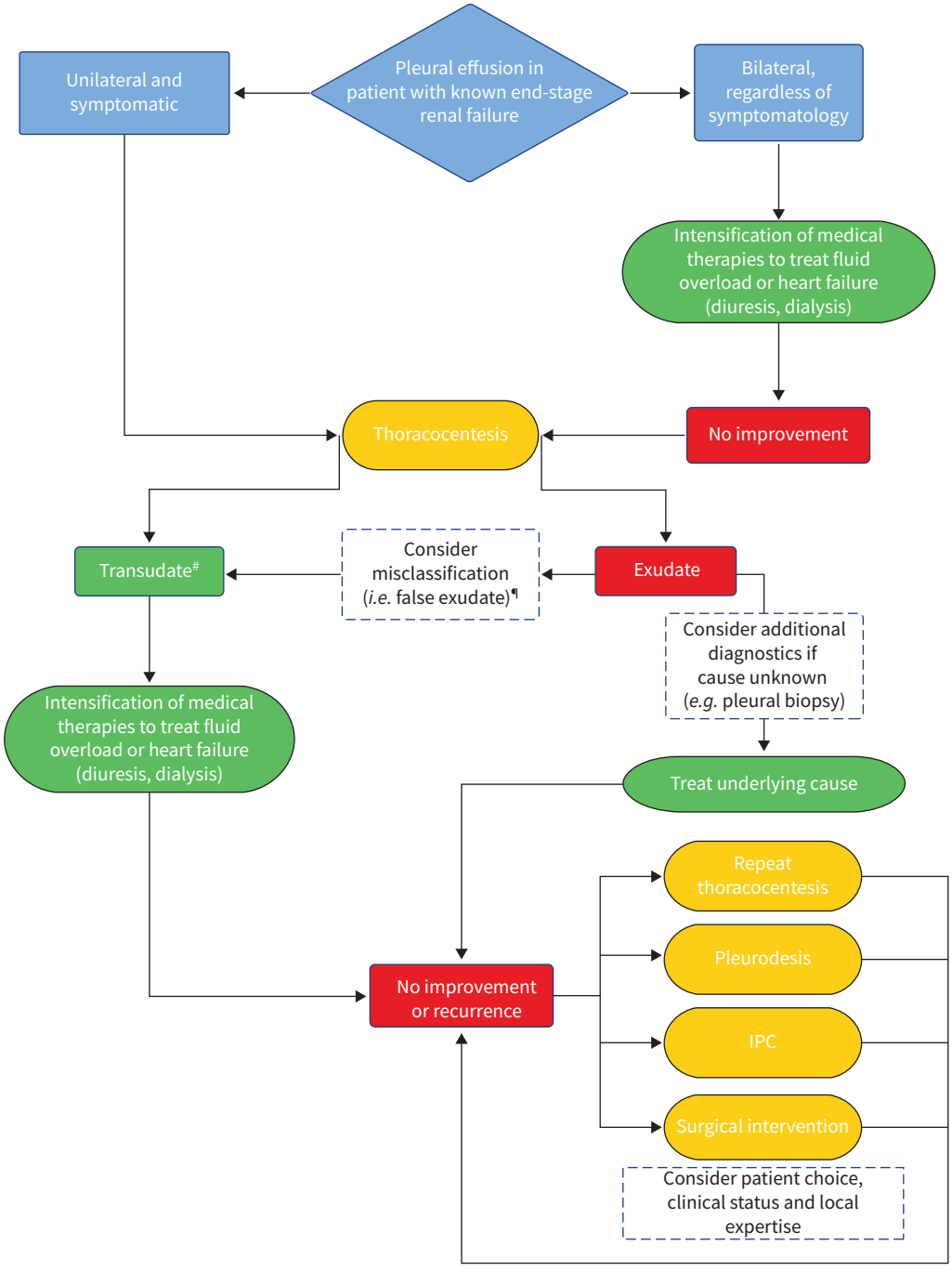


FIGURE 5 A visual depiction of the Task Force members' practice in managing a patient with end-stage renal failure and pleural effusion. This describes the current practice of the Task Force, and is not intended as a recommendation for clinical practice. IPC: indwelling pleural catheter. #: Doelken *et al.* [143] demonstrated that pleural fluid protein content was greater in patients with pleural effusions secondary to end-stage renal failure compared to heart failure (23 $g \cdot L^{-1}$ *versus* 18 $g \cdot L^{-1}$); ¶: Corbett *et al.* [144] demonstrated Light's criteria to be poorly specific (specificity 44%) in the dialysis population, with a greater false positive rate (*i.e.* false exudate) than seen in undifferentiated population.

In one of the few trials of pleural intervention in ESRF patients with recurrent effusions, Potechin *et al.* [148] describe a case series of nine IPCs inserted in eight patients who had recurrent effusion (at least two thoracocenteses in the preceding 2 weeks) and were already on maximal medical therapy (furosemide 160 $mg \cdot day^{-1}$, spironolactone 400 $mg \cdot day^{-1}$) and intolerant of dialysis. The study's primary outcome

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

measure was patient-reported dyspnoea scores, using a baseline dyspnoea index and transitional dyspnoea index (TDI) 2 weeks after IPC insertion. They observed a significant improvement in dyspnoea (median TDI 6) and no significant fall in serum albumin after a median of 34 days post-IPC insertion. Auto-pleurodesis was observed in three patients after a median of 77 days. There were no major complications, including pleural infection. The baseline characteristics indicated a somewhat frail and elderly population. In a thoracoscopy study of 10 ESRF patients with unexplained pleural effusion, COLELLA *et al.* [149] demonstrated safety and successful resolution of recurrent effusion in four patients who received a poudrage.

Only six out of 68 patients in the REDUCE trial and 13 out of 350 patients in the meta-analysis of IPCs in NMPEs had had ESRF. Therefore, interpretation and applicability of the observed findings are difficult in this population [56, 60].

Surgical options are limited, particularly given the overall frailty of this population. Data from early literature showed that invasive surgical interventions, such as decortication and lung release in treating fibrothorax due to uraemic pleuritis, and more aggressive drainage strategies to prevent rounded atelectasis improve lung function and clinical course [150–152]. Presentations of fibrothorax are less commonly encountered since the advent of RRT and most pleural physicians are in agreement that rounded atelectasis itself requires no aggressive therapy.

The data on medical management of effusions in ESRF are also elusive; observational data on 1038 PD patients, of whom 82 had a pleural effusion, found that the pleural effusion resolved in 10 patients (12.2%) with intensive hypertonic PD alone, while the remainder required thoracocentesis for both diagnostic and therapeutic purposes. In the series by JABBAR *et al.* [133], 236 out of 280 patients (84%) were "medically" managed, though specific details of therapy are lacking. A summary of studies of intervention in pleural effusions secondary to ESRF is shown in supplementary material: table E.

Areas of future research

Diagnostic accuracy:

- Can we improve the specificity of Light's criteria for the ESRF population in order to avoid mislabelling a transudate (due to fluid overload) as a false exudate?
- Can we produce through expert consensus (*e.g.* Delphi) precise criteria for defining the aetiology for pleural effusions in ESRF patients?

Treatments:

- What are the most effective therapeutic interventions in addressing recurrent effusions in the ESRF population, with a focus on PROMs, QoL metrics and health economic analyses?

What is the usual investigation and management of a peritoneal dialysis-associated pleuro-peritoneal leak

P: Patients receiving PD

I: Investigation: PF analysis according to defined cut-off values, peritoneal injected contrast agents, trial of cessation from PD
Management: Pleural interventions (pleurodesis *via* slurry, poudrage, surgical), renal interventions (permanently cease PD, temporary pause of PD)

C: Not required

O: Diagnostic accuracy, time to diagnosis, efficacy (resolution of pleural effusion), complications

Summary

- Evidence on options for investigating and treating PD-associated pleuro-peritoneal leaks (PPLs) are limited to observational series.
- The ability to pause and resume PD while observing for recurrence has been used as a diagnostic tool in many of the observational series; several proposed investigative pathways in the literature adopt this approach.
- The literature describes many patients successfully treated with conservative measures; however, a significant few went on to require pleural interventions.
- There is little evidence to draw firm conclusions for the different forms of pleural intervention, and the choice usually depends on both patient suitability and access to surgical services.

In total, 411 studies were screened to identify 14 studies relevant to producing this section. These consisted of one editorial, 11 retrospective observational studies and two case series (supplementary material).

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Pleuro-peritoneal leaks in patients receiving peritoneal dialysis

Hydrothorax due to a PPL in patients receiving PD has a reported incidence of 1.0–5.1% [142, 153–155]. They are mostly unilateral with the majority (88%) occurring on the right side [154]. In one series, 50% of cases occurred within the first 30 days of initiation of PD and a further 18% beyond the first year. Some cases occurred as late as up to 8 years following initiation of PD [154]. In this same series, 26% of the cases were asymptomatic (supplementary material: table F).

Diagnosis of pleuro-peritoneal leak

The PF is clear or straw coloured and has traditionally been described as having the characteristics of an extreme transudate with very low protein values of $<1\ g \cdot dL^{-1}$ and very elevated glucose values of $350–450\ mg \cdot dL^{-1}$ (equivalent to $19.4–25\ mmol \cdot L^{-1}$), and often reflects a value between the serum glucose and dialysate glucose levels [156]. Several studies have assessed diagnostic cut-off values for PF glucose and optimal imaging but there are no studies directly comparing the sensitivity or specificity of each test [157–161]. Figure 6 describes the usual diagnostic and therapeutic approach of TF members in PD patients with a suspected PPL.

Treatment of pleuro-peritoneal leak

Immediate management of a PPL is to usually discontinue PD and consider thoracocentesis for symptom relief in those presenting with dyspnoea. Long-term management strategies for PPL include a retrial of PD after temporary PD cessation (during which alternative RRT may be needed), followed by chemical pleurodesis *via* tube thoracostomy or MT, VATS-assisted mechanical or chemical pleurodesis with or without surgical repair, or finally open surgical repair of any anatomical defects. There are no studies directly comparing the outcomes of these different management approaches.

In one series, hydrothorax resolved completely in 27 (54%) of 50 patients, of whom 19 (38%) had only brief cessation of PD and a further eight patients had low volume exchanges in a semi-erect position or pleurodesis with tetracycline or other agents [154]. In a systematic review by Chow *et al.* [153], temporary discontinuation of PD for a period of 2–6 weeks was successful in 53% of patients without recurrence. Alternatively, low volume PD exchanges in a reclining position have also been used successfully because the reduced intra-abdominal pressure and gravitational factors allow spontaneous closure of the defects. Because there is no reliable way of predicting response to a temporary suspension of PD, other treatment strategies are sometimes considered in addition to this, depending on patient and clinician preference. Conventional tube thoracostomy-directed pleurodesis has been used successfully in 48% of cases [153].

Recurrence following a temporary discontinuation of PD or following pleurodesis has been attributed to large diaphragmatic defects that require surgical repair either by thoracotomy or VATS, with reported success rates of 100% and 88%, respectively [153, 160, 162]. Owing to the perioperative risk associated with thoracotomy, VATS with chemical or mechanical pleurodesis with Marlex or prolene mesh have been increasingly used [153, 163]. In addition to pleurodesis, additional procedures at thoracoscopy such as endoscopic suturing and repair using a Teflon patch have been described to prevent recurrences. The timing of when to resume PD is arbitrary, but a rest period of 3–4 weeks after surgical repair or pleurodesis has been recommended before reinitiating [153]. In one study of 27 patients, by 26 months, withdrawal from PD owing to recurrence was seen in 88% of those managed non-surgically compared to a failure rate of zero in the surgical group (VATS diaphragmatic repair) [160].

Chen *et al.* [164] reported outcomes on 35 patients with PPL who underwent VATS-assisted surgical interventions using a variety of techniques, *e.g.* mechanical pleurodesis with prolene mesh, chemical pleurodesis with talc or tetracycline, direct surgical closure or a combination of the above. The lowest recurrence rates were seen when combination mechanical and talc pleurodesis were performed (10%, compared to 33% for other techniques). Using data from historical cohorts from other studies, they reported that combined pleurodesis had an odds ratio (OR) for recurrence of 0.12 while other techniques had an OR of 0.54 in comparison to discontinuation of PD alone [164]. The mode of delivery of pleurodesis depends upon patient suitability, access to thoracic surgery and the suspicion of sizeable diaphragmatic defects such that more invasive surgical interventions may be warranted [153].

There are no prognostic markers to predict response to treatment with any of the above interventions. However, treatment failure was more common among female patients, patients with polycystic kidney disease and hydrothorax secondary to early leaks (<30 days of commencing PD) [153]. Supplementary material: table G summarises the success rates of various treatments to address a PPL.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

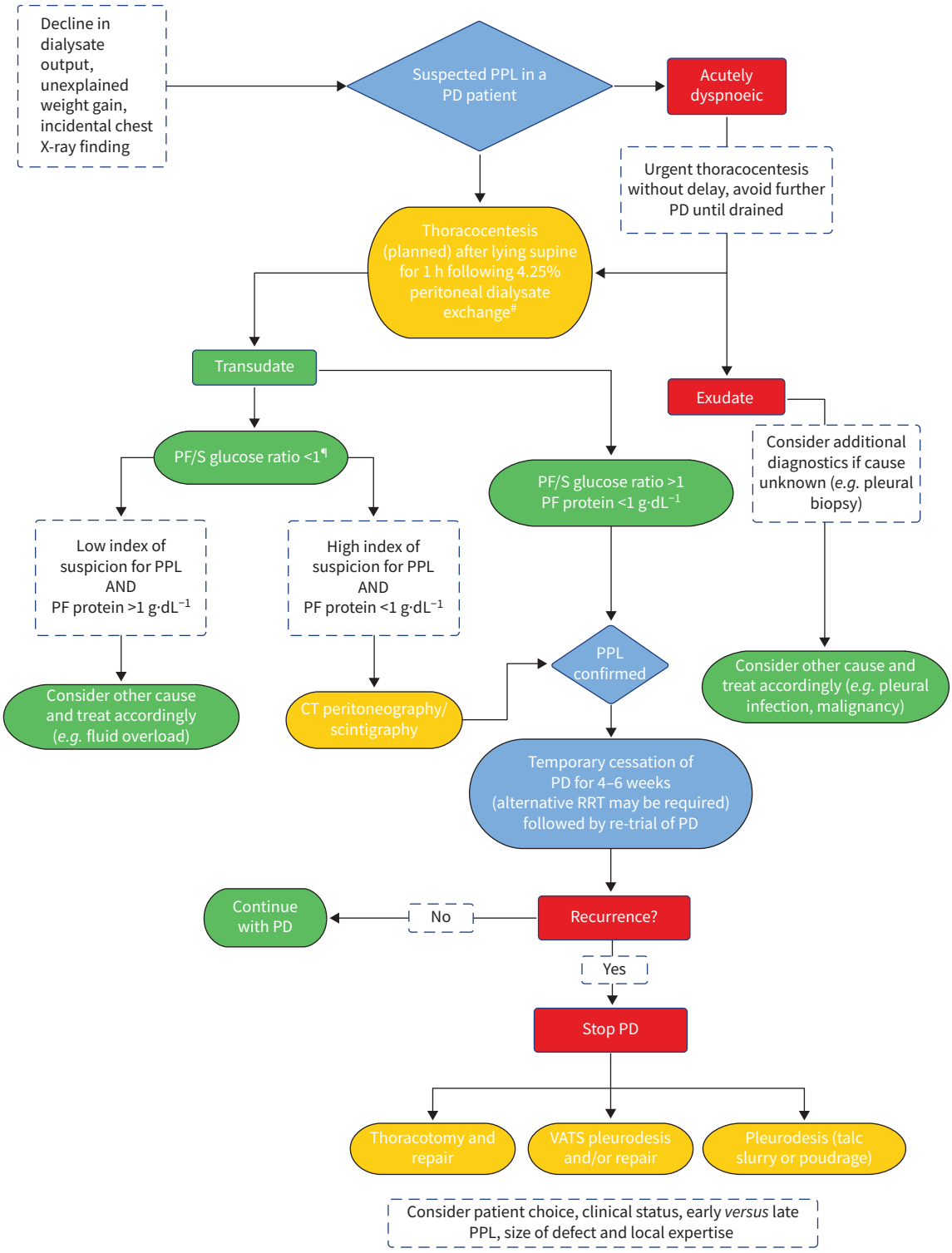


FIGURE 6 A visual depiction of the Task Force members' practice in managing a suspected pleuro-peritoneal leak (PPL) in a peritoneal dialysis (PD) patient. This describes the current practice of the Task Force, and is not intended as a recommendation for clinical practice. PF: pleural fluid; S: serum; CT: computed tomography; RRT: renal replacement therapy. #: Momenin *et al.* [158] proposed a standardised approach to performing thoracocentesis to minimise confounders observed in the PF/S glucose gradient (the size of the diaphragmatic defect and patient posture affecting the rate of fluid movement into the pleural cavity, fluctuating blood sugar levels in patients with diabetes and reabsorption of glucose from the pleural cavity due to delays in obtaining the PF sample); ¶: in the same study across 48 patients with a PPL, 20% of patients had a PF/S glucose gradient <50 mg·dL^{-1} (<2.8 mmol·L^{-1}), 13% had values of 51–100 mg·dL^{-1} (2.8–5.6 mmol·L^{-1}) and 67% had values >100 mg·dL^{-1} (>5.6 mmol·L^{-1}) while the PF/S glucose ratio was always >1.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Areas of future research

Predictors of treatment failure:

- Can we predict which patients with a PD-associated PPL will respond to PD interruption, and offer upfront definitive interventions for this cohort?

Diagnostics:

- Can we define a standardised method for obtaining a PF to serum glucose gradient and/or ratio?
- What form of imaging is most sensitive/specific at diagnosing a PPL?

Treatments:

- What are the most effective interventions to treat a PPL?

Benign asbestos-related pleural effusions

In patients with suspected benign asbestos-related pleural effusions/diffuse pleural thickening, what are the clinical features that can identify risk factors, diagnose the condition and identify prognostic features?

P: Patients with suspected benign asbestos-related pleural effusions (BAPE)/diffuse pleural thickening (DPT)
I: Clinical features (PF, histopathology, radiology), risk factors, prognostic factors
C: Not required
O: Diagnosis of BAPE/DPT/exclusion of mesothelioma, diagnostic accuracy rates of BAPE, rates of evolution of BAPE, rates of lung function decline, rates of radiological progression

Summary

- BAPE and DPT are manifestations of inflammatory-driven responses to exposure to asbestos fibres and in observational series present following a mean latency period from exposure to onset of 30–38 years.
- The cardinal symptoms reported in these studies are cough, breathlessness, chest pain and influenza-like symptoms.
- Diagnostic workup includes plain film and cross-sectional imaging, with histo- or cytological analysis *via* thoracocentesis or biopsy.
- Positron emission tomography (PET) CT is a useful adjunct in differentiating BAPE/DPT from malignant processes.
- DPT is defined as shouldered pleural thickening measuring >5 cm axially by >8 cm craniocaudally with a minimum thickness of 3 mm.
- PF is predominantly a lymphocytic or eosinophilic exudate and may be haemorrhagic.
- Bilateral disease confers a worse symptomatic prognosis in observational series.

In total, 2048 studies were screened to identify 29 studies relevant to producing this section. These consisted of one editorial or narrative review, 17 retrospective observational studies, four prospective observational study, five case series or single case reports, one systematic review and meta-analysis and one trial protocol (supplementary material).

Risk factors

Both BAPE and DPT are known to develop following asbestos exposure [165]. However, there are a number of other causes of DPT that need to be excluded before a firm diagnosis of asbestos-related DPT can be made. This statement will focus on DPT related to asbestos exposure.

BAPE/DPT can occur after low-dose asbestos exposure [166] but is more commonly seen following moderate to high exposures. Its incidence increases with higher exposures in keeping with a dose-response effect [165–168]. DPT has been postulated to occur both as a sequela to an inflammatory BAPE effusion and independently as a result of parenchymal fibrotic inflammation extending into the visceral pleura in line with asbestos fibre migration patterns [169–172]. BAPE/DPT have a mean reported latency period of 30–38 years [167, 173, 174]. High rates of smoking have been observed in patients with BAPE/DPT; however, it is unlikely there is a causative relationship [167]. BAPE/DPT have been most frequently reported in patients exposed to crocidolite and chrysotile fibres; however, no statistically significant data exist to define risk conferred by fibre type [175, 176]. Similarly, BAPE/DPT patients tend to be male, which correlates to occupation and exposure patterns rather than being an independent risk factor [166, 167].

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Clinical features

Patients with BAPE/DPT present to medical services at varying points in their natural disease course. At the point of presentation there is wide variation in symptomatology and some are asymptomatic; the cardinal symptoms are cough, breathlessness, chest pain and influenza-like symptoms [165, 167, 174, 177, 178]. It is expected that there is evidence of asbestos exposure, which may be evident in the clinical history and/or the presence of pleural plaques on chest imaging [179].

Diagnostic investigations

Pertinent diagnostic investigations undertaken in the first instance include CT imaging, thoracocentesis and/or biopsy. Diagnosis remains based on the exclusion of other pathologies, with some authors advocating multidisciplinary team discussion and a period of monitoring of at least 24 months without development of radiological features of malignancy.

Radiology

BAPE effusions occur most frequently on the right (69–76%) but may be left-sided or bilateral at presentation [167, 174, 178, 180, 181]. There are often concomitant signs of asbestos exposure, such as pleural plaques, pleural thickening, asbestosis and folded lung/rounded atelectasis [167, 182].

CT appearances of BAPE may be indistinguishable from early-stage malignant pleural mesothelioma (MPM), hence it is a diagnosis of exclusion. However, pleural thickening >1 cm, pleural nodularity, chest wall invasion, involvement of the mediastinal pleura and high grade "pleural irregularities" are strong indicators of MPE or MPM [182–184]. PET-CT may have a limited role in cases of diagnostic uncertainty, despite tissue biopsy, with MPM often exhibiting higher maximum standardised uptake values levels, prompting consideration of a repeat biopsy [185–187]. DPT is classically defined as thickening of the visceral pleura of ⩾3 mm thickness seen on chest X-ray that obliterates the costophrenic angle [188]. However, Lynch *et al.* [189] define DPT as pleural thickening of ⩾3 mm thickness and measuring >5 cm axially and >8 cm craniocaudally, with or without costophrenic angle involvement (figure 7) [168].

CT may show a constellation of pleural thickening, pleuro-parenchymal bands/"crow's feet" and folded lung [190]. Ultra low-dose CT has been shown to be effective in identifying DPT (sensitivity 90.9%, specificity 100%, positive predictive value 100% and negative predictive value 97.8%). However, intravenous contrast may be of use when excluding malignancy [191]. As with BAPE, PET-CT may be used to identify malignant features [185–187, 192].

Laboratory investigations

PF is usually exudative and often haemorrhagic. It has variable cytological predominance but most commonly is lymphocytic or eosinophilic [3, 9, 11, 30]. By definition, there must be no malignant cells

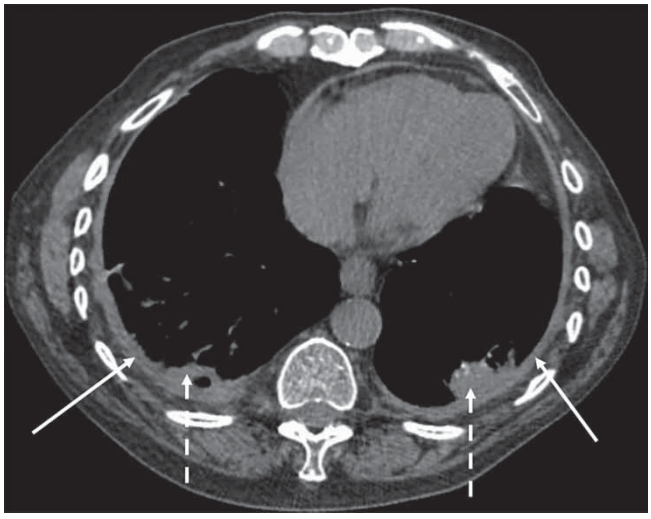


FIGURE 7 Axial computed tomography scan showing typical features of diffuse pleural thickening (solid arrows) with associated folded lung (dashed arrows).

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

present. PF hyaluronic acid and secretory leukocyte peptidase inhibitor levels have been found to be lower in BAPE than malignant processes, thus may be of use in differentiating BAPE from MPM [32, 33].

Investigations for other pathologies such as infection or multisystem inflammatory disease are frequently undertaken to exclude these conditions [3, 34].

Histopathological samples may show a variety of features, including signs of chronic inflammation, the fusion of parietal and visceral pleura, and the loss of submesothelial elastic tissue. Honeycombing and basket weave fibrous focal lymphocytic collections may be seen in lung adjacent to fibrotic pleura. Evidence of chronic asbestos exposure may also be seen in histopathological samples with high asbestos fibre counts [176]. Again, malignant cells must be absent on full thickness biopsies (*i.e.* to fat), as must histopathological markers of mesothelioma such as BRCA1-associated protein 1 (BAP1) loss and p16 [193]. Recent advances in microRNA analysis have shown promise in differentiating between BAPE and MPM but are yet to be developed into validated clinical markers [194].

Clinical course and prognosis

The disease course of BAPE is varied; however, the majority of patients experience a transient pleural effusion that resolves within 3–12 months. Patients may develop contralateral effusions and recurrence on either side [165, 166, 194]. Effusions that do not resolve may become organised over time. BAPE progresses to DPT in 30–40% of cases [174, 179]. The majority of patients with DPT have unilateral disease at presentation; however, 24–39% progress to bilateral disease within 2 years [174, 194]. In those patients whose disease does not progress, DPT exhibits radiological stability. No risk factors or prognostic indicators for progression of BAPE or DPT have been established. Bilateral DPT at presentation and the presence of concomitant asbestosis have been shown to be markers of greater respiratory compromise [194].

Development of malignant pleural mesothelioma

BAPE/DPT have been increasingly recognised to have the potential to evolve to MPM, with a recent meta-analysis finding an evolution rate of 6% and a retrospective study demonstrating 14.5% evolution [195, 196]. The results of a prospective study of BAPE–MPM evolution rates are awaited [197].

Lung function

Both BAPE and DPT are thought to exert a restrictive defect on lung function. In one retrospective series of patients who experienced regression or resolution of effusions, once vital capacity dropped to <60% of expected, it did not recover [179]. This may not directly be due to the presence of effusion, because other studies have shown an only modest impact on lung function due to pleural effusion, and may speak to other mechanisms implicated in BAPE and DPT [81]. DPT has been shown in multiple studies to decrease total lung capacity and diffusing capacity of the lung for carbon monoxide (D_{LCO}) with forced expiratory volume in 1 s/forced vital capacity (FVC) ratio preservation [169, 198, 199]. DPT CT findings are known to correlate with lung function deficit, with bilateral DPT and DPT obliterating the costophrenic angle shown to exert the most potent effect on FVC and D_{LCO} [17, 43]. No correlation has been found between Medical Research Council (MRC) breathlessness scoring and FVC/D_{LCO} or radiological findings, perhaps reflecting the complexity of breathlessness is this cohort [198].

Unfortunately, data on progression of lung function are lacking.

Areas of future research

Diagnostics:

- Can reliable serological or histological biomarkers for BAPE/DPT be identified?
- In the case of negative or unachievable biopsy, can PET-CT be used to reliably differentiate between malignant and non-malignant DPT?
- Can a diagnostic classification system be developed to improve upon current diagnosis of exclusion approach?

Prognostication:

- What is the impact of BAPE or DPT on lung function over time?

In patients with established benign asbestos-related pleural effusions/diffuse pleural thickening, what are the options for follow-up?

P: Patients with established BAPE/DPT
I: Duration of CT follow-up period, use of PET-CT, use of biopsy and/or re-biopsy
C: Not required

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

O: Reduction in length of follow-up and number of interval CT scans, rates of malignancy

Summary

- In the authors' practice, diagnosis is made following biopsy-proven exclusion of a malignant process, with serial follow-up imaging±repeat sampling suggested, over a 2-year period.
- In observational series, the majority of BAPE was seen to resolve over 1 year, although up to 40% developed DPT following effusion resolution.
- BAPE/DPT progresses to mesothelioma in 6–14% of cases; no prognostic indicators of progression to mesothelioma have yet been identified. In the event of suspected progression, re-biopsy may be indicated.

Published evidence for this question is limited. This statement is based upon a synopsis of available literature and the authors' clinical expertise. It is hoped that identifying this area as lacking in evidence will drive future research and investigation.

Biopsy and re-biopsy

Because BAPE/DPT is a diagnosis of exclusion, biopsy is an essential part of workup and, where possible, is usually undertaken at presentation [167, 176, 178]. The biopsy site is guided by imaging and may be undertaken percutaneously, thoracoscopically or surgically (please refer to section on "Nonspecific pleuritis").

In some cases, biopsy will not be possible owing to the lack of an accessible target or poor performance status. There is a dearth of literature to guide management in cases such as these; however, the authors would undertake serial imaging and clinical review to monitor progress. This strategy allows for identification of biopsy targets if they evolve, or the demonstration of clinical stability.

In patients with initial investigations that are suggestive of benign asbestos-related pleural disease, the development of new or evolving pleural lesions with malignant-appearing features would prompt the TF members to always investigate with (re-)biopsy to exclude malignancy, when clinically appropriate [180].

CT follow-up

Patients with BAPE/DPT undergo a period of follow-up, with or without initial biopsy, to ensure resolution or stability of their disease and allow for diagnosis of evolution of malignancy [200]. However, the follow-up period reported in literature varies from months to 30 years [167, 174, 175, 194]. The median time of DPT progression is reported as 2–3 years, which is reflected in the 2-year follow-up period adopted by multiple authorities [174, 178, 180]. Similarly, the follow-up period for malignant evolution currently under investigation in the Meso-ORIGINS study is 2 years [197].

Within the 2-year follow-up, serial CT imaging can demonstrate progression or highlight new changes suggestive of malignancy and may be performed at 6-, 12- and 24-month intervals [174, 178, 180].

PET-CT has been used to effectively differentiate between malignant and non-malignant pleural disease [185, 187, 192]. Kramer *et al.* [192] demonstrated that PET-CT at the point of diagnosis has a negative predictive value for malignant pleural thickening of 0.92 (95% CI 0.78–1.00). Although this study is small, 3 years of follow-up data were included, in which time only one false-negative case (a slow-growing solitary fibrous tumour) was identified. Yildirim *et al.* [187] demonstrated that PET-CT may achieve a sensitivity of 88.2% and specificity of 92.9% for malignant pleural disease, albeit in a smaller sample without a long-term follow-up period. Both studies suggest that PET-CT may be effective in differentiating between malignant and non-malignant pleural disease; however, further prospective data are required before this approach is widely adopted.

Further prospective work is required to delineate the optimum follow-up imaging strategy for BAPE/DPT. Although PET-CT at the point of presentation shows promising diagnostic potential, the reported evolution rates of MPM from larger scale studies indicate serial CT imaging over an extended period might be needed. Therefore, the role of PET-CT currently appears to be in differentiating between malignant and non-malignant disease in equivocal cases and cases where no cytological or histological samples may be obtained, and for identification of areas of high avidity that may represent the most diagnostic biopsy targets.

Areas of future research

Diagnostics:

- Can PET-CT obviate/reduce the requirement for serial CT imaging?

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Prognostication:

- Can prognostic biomarkers of progression from BAPE to DPT and from BAPE/DPT to MPM be identified?

Post-surgical pleural effusions

Usual treatment of post-surgical pleural effusions: thoracocentesis/chest drain versus *conservative treatment*

P: Adults (>18 years old) with post-surgical pleural effusions after cardiac or thoracic surgery

I: Effusion drainage by thoracocentesis or chest drain insertion

C: Medical management or observation

O: Readmission, length of stay, symptom scores, recurrence of effusion, radiological improvement, complications, QoL, mortality

Summary

- Postoperative pleural effusions are common but there is no evidence to suggest they have an impact on morbidity or mortality.
- Aetiology and presentation vary according to primary diagnosis and surgery type and management varies accordingly.
- Studies have shown most effusions do not require intervention and radiological features alone should not dictate the need for intervention. Where studies exist, a protocolised pathway for intervention resulted in small improvements in walking distance, had no impact on QoL or self-reported symptoms, and reduced lengths of hospital stay.
- Most studies on pleural effusion management post-thoracic surgery have focused on timing of postoperative drain removal; earlier drain removal following thoracic surgery, at higher than traditionally accepted drain outputs (450 $mL \cdot day^{-1}$), is safe and efficacious.
- Ultrasound-guided thoracocentesis has largely replaced the need for more invasive surgical tube thoracostomy and is both effective and well tolerated by patients in observational series.
- The literature on "late" postoperative pleural effusions is sparse, and in most instances, comprehensive investigation is carried out before attributing it to a postoperative cause.

Pleural effusions are a nonspecific and well-known complication following cardiothoracic and heart/lung transplant procedures and often follow a benign course. Currently, there is little consensus on the optimal treatment of a postoperative pleural effusion. In total, 900 studies were screened to identify 20 studies relevant to producing this section. These consisted of three editorials or narrative reviews, eight retrospective observational studies, five prospective observational studies, one non-randomised comparative study and three RCTs (supplementary material).

Prevalence of postoperative pleural effusions

Pulmonary complications after cardiothoracic surgery lead to higher mortality rates [201, 202]. While pleural effusions are included among the spectrum of pulmonary complications, to date no evidence has demonstrated an isolated effect of postoperative pleural effusion on mortality.

Postoperative pleural effusions are divided into "early" and "late" categories depending on the time of onset following surgery. Early effusions are those occurring within 30 days of surgery and late beyond 30 days.

Analysis of early pleural effusions show they have higher erythrocyte, LDH and eosinophil counts, whereas late pleural effusions are predominantly lymphocytic, with lower LDH levels. While both are exudative, the differences are likely due to changing biochemical processes occurring at different stages of the postoperative period. Early effusions are more often related to the trauma and bleeding of surgery itself, while the biochemical characteristics of late effusions suggests an immune-mediated response [203]. Pleural effusions may also arise as a result of operative damage to the thoracic duct (chylothorax) or represent a postoperative infection.

Several authors classify pleural effusions into "clinically significant" and "clinically nonsignificant" based on symptoms: increased respiratory support, shortness of breath, cough, tachypnoea and pain. While a large proportion of patients (42–89%) have radiographic findings of pleural effusion in the early postoperative period, not all of them are considered clinically significant requiring intervention [201, 203–205]. Diagnosis of and intervention in late pleural effusions can be somewhat mixed, because these vary according to the usual follow-up pathways among different surgical units and there is a paucity of literature on their characteristics or management.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

The prevalence of pleural effusions requiring intervention following coronary artery bypass grafting (CABG) or valve surgery is ~6.6% [203]. Large or symptomatic pleural effusions usually prompt intervention (thoracocentesis or chest drain), with the definition of a large effusion differing: >25%, >30% or >33% of the hemithorax on a frontal chest radiograph in various studies [206–208]. Decisions to intervene are usually based on a combination of clinical and radiological features rather than individual parameters [209]. Xing *et al.* [210] report that 1.2% patients require thoracocentesis owing to recurrent pleural effusions, but do not necessarily specify what criteria prompted intervention in some and not others. Moreover, recurrence despite intervention is documented in around 21% [203].

In an observational series, Usta *et al.* [209] suggested that a protocolised pathway for intervention in postoperative pleural effusion (if symptomatic and estimated pleural effusion volume >480 mL) reduced length of stay by 3±1.5 days compared to patients managed with diuresis alone. Hansen *et al.* [211] investigated whether dedicated follow-up and intervention on pleural effusions enhanced recovery rates or improved QoL and symptoms. They compared a standard follow-up protocol with physicians' decision on when to intervene on pleural effusions with more frequent follow-up with a dedicated protocol for intervention (effusions of estimated volume >400 mL or <400 mL and symptomatic). They found no significant difference in self-reported symptoms or QoL, but an improvement in walking distance when the pleural effusions were drained according to a standardised protocol. They conclude that recovery rates can be enhanced by up to 15% with dedicated follow-up and drainage regimes. Ultrasound-guided thoracocentesis has now replaced the more invasive surgical tube thoracostomy as the initial intervention of choice, and is well tolerated by patients [209–211].

Effusion after cardiac surgery

There are differences in the presentation and management of postoperative effusions according to surgery type: cardiac *versus* thoracic. This can in part be explained by the differences in comorbidity, which have led to surgical intervention in the first instance, and differences in surgical technique and approach. There is more evidence in the medical literature on pleural effusion management following cardiac surgery (*e.g.* CABG, valve replacements and aortic interventions) than following thoracic surgery (*e.g.* lung resection). A further distinct entity, post-pericardiotomy syndrome, is characterised by fever, pleuritic pain and pleural and/or pericardial effusion and is thought to represent an auto-inflammatory process [212, 213]. In symptomatic patients, anti-inflammatories (nonsteroidal anti-inflammatory drugs, aspirin, colchicine and glucocorticoids) may be beneficial [213]. However, only postoperative colchicine has been shown to be of benefit as a preventative measure in reducing the incidence of post-pericardiotomy syndrome [214].

Differences between techniques within the same procedure have been studied. CABG with internal mammary artery harvesting and preserving the integrity of the pleura had a lower rate of pleural effusion development compared to internal mammary artery harvesting and breaching of the pleura [215]. One study addressed pulmonary compromise after a maze procedure, which entails creating scar tissue in the atria with a scalpel or an energy-delivering device to stop abnormal electrical conduction and thus treat atrial fibrillation. Ad *et al.* [216] compared prophylactic continuous furosemide infusion with boluses after the maze procedure and demonstrated that the former had a significantly lower incidence of pulmonary complications and need for effusion drainage. However, the maze procedure is thought to reduce the production of atrial natriuretic peptide, which may underpin a different mechanism in the development of effusion, compared to more routine cardiac surgical procedures.

Effusion after thoracic surgery

Literature on postoperative effusion management following thoracic surgery has concentrated more on the optimal timing of chest drain removal and the impact on important outcome measures such as readmission rates and need for further pleural intervention. Cut-offs of the ideal amount of drainage per day before removal of the chest drain are based on expert opinion rather than evidence; hence, several studies have attempted to assess greater than normally accepted practice limits (250 mL·day^{-1}) [217]. Higher thresholds (450 mL·day^{-1} *versus* 250 mL·day^{-1}) for chest tube removal post-lung resection improved respiratory function, reduced infection rate and pain symptoms, and enabled chest tubes to be taken out earlier. Relatively low effusion recurrence and re-intervention rates were observed. Earlier chest tube removal reduced the length of stay in hospital [210, 217]. Readmission rates due to symptomatic pleural effusion were low and did not differ between higher and lower threshold groups [217]. Interventions following readmission due to symptomatic effusions varied between repeated VATS/thoracotomy, thoracocentesis and observation [217]. A Danish study by Bjerregaard *et al.* [218] noted that only 2.8% of patients developed pleural effusions requiring re-intervention, despite allowing high volume output before removing the chest tube, and most patients with recurrence could be managed without invasive intervention.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Overall, these studies show that higher drainage thresholds of up to 450 mL·day^{-1} used for chest drain removal is safe, with low re-intervention rates and improved patient experience [210, 217, 218].

Effusion post-transplantation

Pleural effusions are a common occurrence in lung transplant recipients, reported in 25–100% of patients in the early period [219, 220]. These effusions are usually exudative, tend to be bloody, have predominant neutrophilia and are usually small to moderate in size [220]. Several causes of effusions during the early postoperative period have been suggested, including pleural inflammation, increased alveolar permeability, atelectasis, impaired lymphatic drainage and host immune response [220]. Late effusions (>15 days post-transplant) are common and are usually exudative and lymphocyte-predominant and tend not to recur following thoracocentesis.

Although post-orthotopic heart transplant pleural effusions have been described in several studies, few data exist describing the characteristics of these effusions [202, 221]. Early post-orthotopic heart transplant effusions are usually bilateral, exudative, moderate to large and usually more neutrophilic and less haemorrhagic compared to post-CABG effusions [205].

Limitations

No RCTs comparing interventions against conservative measures for recurrent postoperative pleural effusions exist. Alternative options for patients who did not receive thoracocentesis for recurrent pleural effusions are described in the literature but the specific details of therapies are vague, and these were not head-to-head comparisons.

Owing to the different nature of cardiac and thoracic surgery, it is difficult to compare the re-intervention rates between the two. Management of postoperative drains (and timing of removal) differs between these two groups; this may also impact the rates of recurrence and decisions for subsequent re-intervention between groups.

Areas of future research

Diagnosis:

- Can we establish diagnostic criteria for late postoperative pleural effusion?

Management:

- Is thoracocentesis more effective than medical management (*i.e.* diuresis) in post-cardiac surgery-related pleural effusions?

Nonspecific pleuritis

What are the options for investigating a patient with an initial nonspecific pleuritis finding on histology?

P: Patients with NSP on histology from image-guided pleural biopsies and/or thoracoscopic pleural biopsies
I: Further biopsy procedures: MT, image-guided biopsies±VATS, conservative management/observation, further ancillary tests (biomarkers/histological techniques)
C: Not required
O: Detection of underlying malignant process (*i.e.* false negative), identification of additional pathologies

Summary

- A histological diagnosis of NSP following thoracoscopy has been found in up to 40% of cases in various series and the most difficult question is whether this is truly a benign diagnosis or a false negative. A possible malignant aetiology has been found in 8% of patients, with mesothelioma the most common malignancy diagnosed.
- Histological patterns of inflammation and fibrosis vary greatly in individual cases of NSP, and likely represent different stages in the process of tissue injury and remodelling. Ancillary techniques, such as fluorescent *in situ* hybridisation (FISH) for cyclin-dependent kinase inhibitor 2A (CDKN2A) deletion and/or BAP1 or 50-methylthioadenosine (MTAP) immunohistochemical loss are essential in differentiating between NSP and mesothelioma *in situ*. Clinical features that may suggest a false-negative NSP result mirror those that make one suspicious for malignancy (particularly MPM).
- Although there is evidence for both TUS and DWI in distinguishing benign from malignant pleural disease, most of the evidence for differentiating malignant from benign pleural disease comes from CT studies.
- Biomarkers for identification of NSP are not well established and their role remains unclear.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

- Parietal pleural biopsy using rigid forceps has a better diagnostic yield compared to biopsy with flexible forceps or cryobiopsy. Adipose tissue is more likely to be present using rigid forceps. Given that invasion of adipose layers is a crucial diagnostic feature for mesothelioma, this is a significant advantage of rigid forceps biopsy.
- In the authors' experience, false-negative diagnoses of NSP, which eventually turn out to be mesothelioma, may be related to difficulties in visualising the pleural cavity at thoracoscopy due to adhesions, excessive fibrinous thickening or early-stage malignant disease with no obvious target for biopsy, rather than type of thoracoscopy performed.

In total, 1949 studies were screened to identify 40 studies relevant to producing this section. These consisted of five editorials or narrative reviews, 22 retrospective observational studies, three prospective observational studies, three RCTs, two laboratory studies, four systematic reviews and meta-analyses and one guideline (supplementary material).

Definition and incidence of nonspecific pleuritis

Histological analysis of a thoracoscopic pleural biopsy is considered the "gold standard" for diagnosis in pleural disease and is recommended as the final diagnostic step for an unexplained exudative effusion [222]. However, in up to 40% of pleural biopsies, a histological pattern of NSP is found [223–237]. In contrast to malignant and tuberculous pleuritis, which have distinct histological features, most other causes of pleural disease are associated with nonspecific biopsy findings (inflammation and fibrosis) [238]. There are currently no recommendations regarding the optimal diagnostic workup in patients with a histological diagnosis of NSP. The TF members include in their baseline evaluation a detailed history, including history of asbestos exposure, and a panel of blood tests and imaging modalities according to pre-test suspicion of aetiology. Following workup, a probable cause is identified in most cases, but in up to 48%, a clear cause cannot be identified and a clinical diagnosis of "idiopathic pleuritis" is made [223–230, 232, 233, 235, 236, 238–240].

While most pleural effusions associated with biopsy findings of NSP resolve spontaneously, in 8% (range 3–38%) a malignant aetiology is eventually recognised, and MPM is the most frequent cause [223, 225–227, 229, 230, 232, 233, 235–242]. The substantial range in the eventual development of pleural malignancy following a finding of NSP may be due to several factors: inter- and intra-centre differences in thoracoscopy operator experience and technique, pathology technique and expertise, geographical variation in prevalence of asbestos exposure and follow-periods between studies. When a diagnosis of NSP is made, the first concern is whether it represents a truly benign condition or a false-negative biopsy due to sampling or other diagnostic error. To date, there is no consensus on what constitutes a false-negative result due to error, as opposed to late progression to malignancy [223].

Histopathological features of nonspecific pleuritis

Microscopic examination of these cases reveals varying degrees of inflammation and fibrosis (figure 8).

An explanation for the wide range of eventual malignant diagnoses following an NSP finding across different datasets may be due to pathology expertise. Use of ancillary histological techniques may minimise false-negatives; the best evidence tests to distinguish MPM from reactive mesothelial conditions are homozygous deletion of *CDKN2A* detected by FISH (100% specificity, 14–85% sensitivity) and loss of BAP1 expression on immunohistochemistry (IHC) (high specificity, lower sensitivity) [243]. Loss of MTAP on IHC has recently been shown to be useful; in a study of 125 MPMs diagnosed at the National MESOPATH Reference Centre, loss of MTAP cytoplasmic staining showed 96% specificity and 86% sensitivity to detect homozygous deletion of *CDKN2A* [244, 245].

Time to progression from mesothelioma *in situ* to invasive MPM has been shown to range from 1 to 15 years [246]. Therefore, the latest World Health Organization classification system for MPM acknowledges "mesothelioma *in situ*" as a stage of disease and defines it as: non-resolving pleural effusion, no thoracoscopic or imaging evidence of tumour, single layer (but papillary proliferations are accepted) of mesothelial cells on the pleural surface without invasion, and loss of BAP1 or MTAP by IHC, or *CDKN2A* homozygous deletion by FISH, followed by multidisciplinary discussion [247]. It therefore remains unclear which cases should undergo further histological analysis with these techniques [247, 248]. The authors' practice is to use them in all cases of suspected MPM.

Factors associated with a false-negative finding of nonspecific pleuritis on biopsy

Clinical features

Davies *et al.* [233] found a strong association between the referring physicians' level of clinical suspicion of pleural malignancy prior to thoracoscopy and eventual malignant diagnosis following an NSP result, but

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

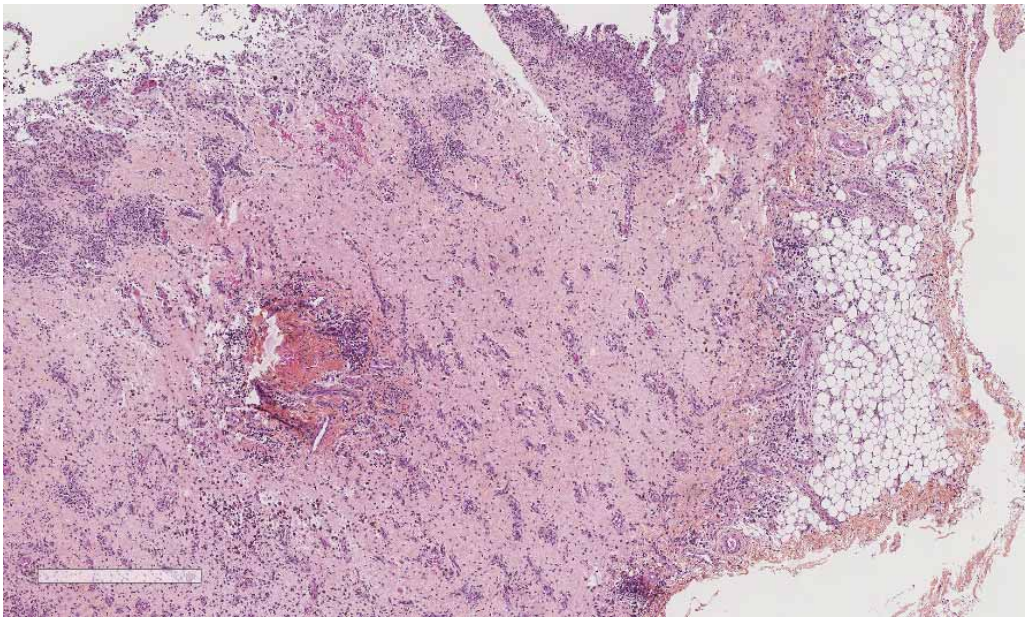


FIGURE 8 A severely thickened parietal pleura consisting of fibrosis, inflammatory elements and numerous blood vessels without any granulomatous or malignant process (haematoxylin, eosin, safran colouration ×40). Scale bar: 600 μm.

found the operator's impression at time of thoracoscopy to be a poor predictor. They also found recurrence of pleural effusion associated with eventual malignant diagnosis [233]. This was also noted by Yu *et al.* [236], who reported a greater likelihood of diagnosing pleural malignancies in patients with an initial false-negative NSP result when faced with recurrent pleural effusion, pleural nodules or pleural plaques.

Clinical risk factors for a false-negative NSP result mirror those for suspicion of malignancy (particularly MPM). Factors that may warrant heightened surveillance and further biopsy include recurrence of pleural effusion, ipsilateral chest pain and history of asbestos exposure.

Imaging features

When NSP is identified in pleural biopsies after thoracoscopy, imaging findings may lead the clinician to suspect malignancy (*i.e.* false-negative NSP) and pursue further workup.

Computed tomography

Across two studies, CT findings suggestive of malignant involvement (parietal pleural thickening (>1 cm), nodular pleural thickening, mediastinal pleural thickening, circumferential pleural thickening and a mass lesion) were associated with a false-negative NSP result [223, 234]. In a recent study, 102 cases of NSP or atypical mesothelial proliferation diagnosed at VATS, progression of CT findings determined whether re-biopsy was attempted [249].

Thoracic ultrasound

Several studies have explored the utility of morphological findings of TUS as a tool for detecting MPE. Certain TUS findings were shown to have a high sensitivity and specificity for MPE [96]. These findings were disputed in a recent meta-analysis, which suggested pleural thickening has low sensitivity and variable specificity, and only visceral or parietal nodularity is specific for malignant pleural disease with a high positive predictive value [52].

Magnetic resonance imaging

There is growing interest in the role of DWI, which can distinguish tumoural from non-tumoural tissue. In a recent study, the sensitivity of DWI in detecting pleural malignancy exceeded that of CT (94.2% *versus* 67.3%), with similar specificity [250]. Revelli *et al.* [251] demonstrated DWI was able to distinguish benign pleural disease from MPM even in small lesions and in the sarcomatoid subtype in patients undergoing thoracoscopic biopsy (n=56).

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

Although both TUS findings and DWI evaluation play a role in distinguishing benign from malignant pleural disease, given the abundance of evidence and minimal inter-operator variability, the most frequently used imaging modality remains CT.

Biomarkers

MPM is the main malignant diagnosis of concern following an NSP histological finding. Biomarkers for early detection of MPM in a high-risk population have been widely studied, and promising biomarkers include mesothelin, osteopontin and fibulin-3. Only mesothelin has received US Food and Drug Administration approval for clinical use [252]. In early mesothelioma diagnosis, mesothelin has a higher sensitivity for MPM in pleural effusion *versus* serum (79% *versus* 61%, with estimates of specificity of 85% and 87%, respectively) [253]. While numerous studies have demonstrated that pleural effusion-soluble mesothelin related peptide (PE-SMRP) levels are higher in patients with MPM than in subjects with benign mesothelial lesions [254], no data are available regarding the role of PE-SMRP as a blanket tool in subjects following a histological finding of NSP.

For the clinician, a biomarker able to identify patients more likely to develop pleural malignancy after an NSP diagnosis is an important tool that might improve management (*e.g.* enhanced surveillance and follow-up). Not all guidelines recommend the use of mesothelin as a diagnostic tool in MPM [255], and further work is required to determine whether it can be used to increase the pre-test probability of an eventual malignant diagnosis following a biopsy finding of NSP [255].

Biopsy technique

The options for obtaining a pleural biopsy in undiagnosed exudative pleural effusions include image-guided pleural biopsies (ultrasound or CT-guided) and thoracoscopic pleural biopsies (MT or VATS) [222].

Image-guided pleural biopsies

Image-guided pleural biopsies have a high sensitivity for the diagnosis of malignancy using both CT and ultrasound guidance [256]. A recent systematic review and meta-analysis including 30 studies showed a pooled yield with both procedures of 84% for TUS-guided and 93% for CT-guided biopsy without meaningful differences in safety [257]. However, diagnostic yield with these techniques can depend on radiological features (*e.g.* identifying pleural thickening) in contrast to direct visualisation of the parietal pleura through thoracoscopy [258]. The authors, therefore, generally reserve image-guided biopsies for those patients in whom thoracoscopy is not possible [222].

Rigid and semi-rigid thoracoscopic biopsies

MT allows direct visualisation of the pleura and allows targeted biopsies, providing a specificity of close to 100% for malignancy, and sensitivity of between 85% and 94% [222].

In 14 studies of 4189 patients with undiagnosed pleural effusion undergoing MT, 1665 received a histological diagnosis of NSP (40%, 95% CI 38–41%). Of the cases where outcomes were reported, 182 out of 2249 cases of malignant pleural pathology were eventually established (8%, 95% CI 7–9%), mainly MPM [223]. Most false-negative cases were attributed to sampling errors; multiple factors bear consideration with regards to biopsy technique and are covered below.

Number of biopsies

While early studies on thoracoscopic technique recommended a greater number of biopsies (*e.g.* 15–20), with advancements in procedural and pathological techniques this has reduced to just 5–10 [224, 235, 259].

Depth of biopsies

A crucial histological feature to make a diagnosis of MPM is adipose tissue invasion. In the presence of thickened pleura or when difficult to obtain deeper biopsies, a “biopsy into biopsy” technique with repeated sampling at the same site has been proposed to achieve “full thickness” biopsies and reduce false-negative sampling errors [260].

Size of biopsy specimens and type of scope

MT can be performed using either rigid or semi-rigid thoracoscopes; this directly determines the size of biopsy specimens obtained [261]. Supplementary material: table H summarises studies comparing different thoracoscopic biopsy techniques.

Biopsy specimens were found to be larger when using rigid *versus* semi-rigid forceps (24±12.9 mm^2 *versus* 11.2±7.6 mm^2) [262]. In patients with extensive adhesions, superior diagnostic yields were attained from

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

rigid compared to semi-rigid forceps (97.8% *versus* 73.3, respectively) [263]. Different strategies have been explored to overcome the limitations of small biopsies acquired with the semi-rigid thoracoscope, including the use of cryobiopsy. A recent systematic review and meta-analysis (22 studies, 1647 biopsies) demonstrated cryobiopsy did not increase diagnostic yield over semi-rigid forceps in semi-rigid thoracoscopy [264, 265]. The only intra-patient comparison of parietal pleural biopsies acquired using rigid and semi-rigid forceps and cryobiopsy during MT confirms the superior yield of rigid forceps biopsy over cryobiopsy (difference: 12.7%). Rigid forceps biopsy specimens were larger than those from cryobiopsy or semi-rigid forceps biopsy, and deep biopsies containing fat were collected in 63% of rigid forceps biopsies *versus* 49.5% and 39.5% in cryobiopsies and semi-rigid forceps biopsies, respectively [266]. There are no studies on the prevalence of NSP histology and eventual malignant diagnoses comparing these techniques.

Video-assisted thoracoscopic surgical biopsies

No RCTs have compared MT and VATS, and only few retrospective studies have evaluated the diagnostic accuracy of both MT and VATS in excluding malignancy.

A retrospective study in 177 patients comparing MT (n=78) and VATS (n=99) showed similar diagnostic yield (93.6% and 96%, respectively). While the incidence of NSP in MT (43.8%) was greater compared to VATS (24.2%), this did not result in differences in rates of eventual malignant diagnosis during follow-up (12.5% MT *versus* 17.4% VATS) [267].

A more recent retrospective study evaluated the eventual malignancy rates in 295 patients initially diagnosed with NSP after MT (1.7%) or VATS (0.55%) and demonstrated good negative predictive values for both (0.982 and 0.994, respectively) [242].

In a further large VATS series (n=400), patients suspected to have MPM underwent biopsy, of whom 102 (25.5%) were deemed to have either NSP or atypical mesothelial proliferation. Over a median follow-up of 5 months, a further 11 (10.7%) underwent repeat biopsy and were subsequently found to have malignancy, giving an overall sensitivity for VATS of 96%. Reasons for misdiagnosis included lack of macroscopic pleural abnormalities, extensive adhesions and dense fibrosis making adequate tissue acquisition challenging [249].

To summarise, pleural biopsies through MT or VATS appear to have similar incidence for NSP histology and a similar negative predictive value in excluding pleural malignancy. An NSP diagnosis made by MT does not therefore suggest the need to repeat the biopsy through VATS. The authors would usually consider a repeat biopsy after MT or VATS in patients found to have DPT with no target area for biopsy, or dense adhesions impeding complete inspection (figure 9); in the latter scenario, the authors sometimes consider a more invasive technique (*e.g.* thoracotomy) to allow complete inspection and biopsy.

Areas of future research

Diagnosis:

- What is the role of ancillary pathology techniques in differentiating NSP from pre-invasive mesothelial lesions?
- How reliable is the thoracoscopist's impression when faced with a finding of NSP?

Prognostication:

- Can we establish a risk stratification model for patients following an NSP result that incorporates clinical data points, imaging, and thoracoscopic and histological findings, and makes use of relevant biomarkers?

What are the options for follow-up in patients with a nonspecific pleuritis diagnosis?

P: Patients with a histologically confirmed diagnosis of NSP
I: Follow-up duration, PET-CT as part of interval imaging
C: Not required
O: Rates and timing of evolving malignancy

Summary

- In the literature, when faced with a false-negative NSP result, an eventual malignant diagnosis is established within a median period of 6 months but may take longer in MPM. It is the usual practice of the authors to follow up patients for 1 year, which may be extended if there are risks factors for MPM.

In total, 105 studies were screened to identify nine studies relevant to producing this section. These consisted of nine retrospective observational studies (supplementary material).

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

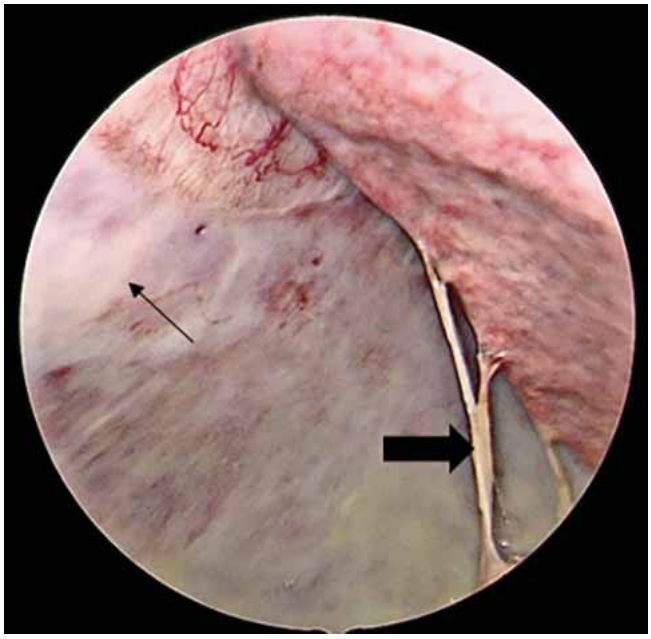


FIGURE 9 Endoscopic appearance of pleural cavity with adhesions (bold arrow) and fibrinous layer on the parietal pleura (thin arrows).

Follow-up duration of patients with a nonspecific pleuritis diagnosis, evidence in the literature

Supplementary material: table I summarises studies of NSP with rates of eventual malignancy and follow-up durations.

In the majority of cases, malignancy was detected within 12 months of initial NSP biopsy [229, 232, 233, 236–242]. Janssen *et al.* [232] found eventual malignancy diagnosed following NSP after a mean follow-up of 4.4 months, though when studying MPM diagnoses specifically, the mean interval was 8.7 months, indicating longer latency. Of note, the studies by DePew *et al.* [239] and Metintas *et al.* [234] found patients eventually diagnosed with malignancy reported persistent pleural pain. Reuter *et al.* [237] developed the concept of number-needed-to-follow-up (NNF) in NSP to diagnose a single case of malignancy, and found an NNF of 18 in year 1, increasing to 260 by year 3.

The TF members usually follow-up patients for a period of 1 year in order to pick up most eventual malignancies, although in the presence of red-flag symptoms (*e.g.* chest pain) or risk factors for MPM, this might be further extended.

Positron emission tomography-computed tomography scan as part of interval imaging

No studies have addressed the use of PET-CT scan in the follow-up of NSP.

Areas of future research

Diagnosis:

- When faced with eventual malignant diagnosis, what time cut-off should be used to declare an NSP result as false negative *versus* a *de novo* malignant process?

Follow-up:

- Is PET-CT more sensitive at detecting the evolution of malignant pleural disease following an NSP result during surveillance follow-up compared to standard of care (*e.g.* CT)?
- What is the role of TUS in NSP follow-up?

Conclusion

This statement has summarised the evidence base on the management of the most pressing causes of NMPEs. A cross-cutting theme across the statement was the need for a multidisciplinary approach. Several

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

of the conditions outlined in the statement require input from single-organ specialty groups, but equally the benefit of a more generalist and holistic perspective cannot be overstated. In particular, the involvement of the palliative care team earlier in the pathway, when deemed appropriate, would serve many of these patients well and provide them with the support needed in the face of a life-limiting condition. NMPEs represent a significant burden of disease and, as this statement has demonstrated, there remains a paucity of evidence to guide best practice. However, this should be viewed as an opportunity and the areas for future research highlighted in this statement represent a starting point to begin addressing these gaps in the evidence base.

This document was endorsed by the ERS Executive Committee on 5 July 2024.

Acknowledgements: The authors would like to acknowledge the support and guidance of the ERS administrative and methodology team, in particular Valérie Vaccaro, Valeria Finelli and Natalia Mikhaylova, and Eihab Bedawi for his encouragement and advice throughout the process.

Conflicts of interest: E.M. Grabczak reports grants from National Science Centre, support for attending meetings from Euromedical, and a leadership role with ERS as secretary of assembly 14. B. Nagavci was the ERS in-house methodologist (2020–2022). The remaining authors have no potential conflicts of interest to disclose.

Support statement: This work was supported by the European Respiratory Society. Funding information for this article has been deposited with the Crossref Funder Registry.

References

1 Bodtger U, Hallifax RJ. Epidemiology: why is pleural disease becoming more common? *In*: Maskell NA, Laursen CB, Lee YCG, eds. Pleural Disease (ERS Monograph). Sheffield, European Respiratory Society, 2020; pp. 1–12.
2 Marel M, Zrůstová M, Stasný B, *et al.* The incidence of pleural effusion in a well-defined region. Epidemiologic study in central Bohemia. *Chest* 1993; 104: 1486–1489.
3 Bintcliffe OJ, Lee GYC, Rahman NM, *et al.* The management of benign non-infective pleural effusions. *Eur Respir Rev* 2016; 25: 303–316.
4 Walker SP, Morley AJ, Stadon L, *et al.* Nonmalignant pleural effusions: a prospective study of 356 consecutive unselected patients. *Chest* 2017; 151: 1099–1105.
5 Mummadi SR, Stoller JK, Lopez R, *et al.* Epidemiology of adult pleural disease in the United States. *Chest* 2021; 160: 1534–1551.
6 Bedawi EO, Ricciardi S, Hassan M, *et al.* ERS/ESTS statement on the management of pleural infection in adults. *Eur Respir J* 2023; 61: 2201062.
7 Feller-Kopman D, Light R. Pleural disease. *N Engl J Med* 2018; 378: 740–751.
8 Porcel JM, Light RW. Diagnostic approach to pleural effusion in adults. *Am Fam Physician* 2006; 73: 1211–1220.
9 Porcel JM, Esquerda A, Vives M, *et al.* Etiology of pleural effusions: analysis of more than 3,000 consecutive thoracenteses. *Arch Bronconeumol* 2014; 50: 161–165.
10 Porcel JM, Light RW. Pleural fluid analysis: are Light's criteria still relevant after half a century? *Clin Chest Med* 2021; 42: 599–609.
11 Porcel JM, Light RW. Pleural effusions. *Dis Mon* 2013; 59: 29–57.
12 Bielsa S, Acosta C, Pardina M, *et al.* Tuberculous pleural effusion: clinical characteristics of 320 patients. *Arch Bronconeumol (Engl Ed)* 2019; 55: 17–22.
13 Porcel JM, Esquerda A, Porcel L, *et al.* Is pulmonary embolism associated with pleural transudates, exudates, or both? *Spanish J Med* 2021; 1: 1.
14 Mercer RM, Varatharajah R, Shepherd G, *et al.* Critical analysis of the utility of initial pleural aspiration in the diagnosis and management of suspected malignant pleural effusion. *BMJ Open Resp Res* 2020; 7: e000701.
15 Porcel J S-M, Bielsa S. Malignant pleural effusions with transudative characteristics. *Gazzetta Medica Italiana* 2022; 181: 482–483.
16 Arnold DT, De Fonseka D, Perry S, *et al.* Investigating unilateral pleural effusions: the role of cytology. *Eur Respir J* 2018; 52: 1801254.
17 Bintcliffe OJ, Hooper CE, Rider IJ, *et al.* Unilateral pleural effusions with more than one apparent etiology. A prospective observational study. *Ann Am Thorac Soc* 2016; 13: 1050–1056.
18 Light RW, Macgregor MI, Luchsinger PC, *et al.* Pleural effusions: the diagnostic separation of transudates and exudates. *Ann Intern Med* 1972; 77: 507–513.
19 Hooper C, Lee GYC, Maskell NA. Investigation of a unilateral pleural effusion in adults: British Thoracic Society pleural disease guideline 2010. *Thorax* 2010; 65: ii4–ii17.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

20 Heffner J. Pleural fluid analysis. *In*: Light RW, Lee YCG, eds. Textbook of Pleural Diseases. 3rd Edn. Boca Raton, CRC Press, 2016; pp. 153–173.
21 Gázquez I, Porcel JM, Vives M, *et al.* Comparative analysis of Light's criteria and other biochemical parameters for distinguishing transudates from exudates. *Respir Med* 1998; 92: 762–765.
22 Porcel JM, Vives M, de Vera MC V, *et al.* Useful tests on pleural fluid that distinguish transudates from exudates. *Ann Clin Biochem* 2001; 38: 671–675.
23 Lépine P-A, Thomas R, Nguyen S, *et al.* Simplified criteria using pleural fluid cholesterol and lactate dehydrogenase to distinguish between exudative and transudative pleural effusions. *Respiration* 2019; 98: 48–54.
24 Heffner JE, Highland K, Brown LK. A meta-analysis derivation of continuous likelihood ratios for diagnosing pleural fluid exudates. *Am J Respir Crit Care Med* 2003; 167: 1591–1599.
25 Porcel JM, Peña JM, Vicente de Vera C, *et al.* Bayesian analysis using continuous likelihood ratios for identifying pleural exudates. *Respir Med* 2006; 100: 1960–1965.
26 Bielsa S, Porcel JM, Castellote J, *et al.* Solving the Light's criteria misclassification rate of cardiac and hepatic transudates. *Respirology* 2012; 17: 721–726.
27 Kolditz M, Halank M, Schiemanck CS, *et al.* High diagnostic accuracy of NT-proBNP for cardiac origin of pleural effusions. *Eur Respir J* 2006; 28: 144–150.
28 Han CH, Choi JE, Chung JH. Clinical utility of pleural fluid NT-pro-brain natriuretic peptide (NT-proBNP) in patients with pleural effusions. *Intern Med* 2008; 47: 1669–1674.
29 Porcel JM, Martínez-Alonso M, Cao G, *et al.* Biomarkers of heart failure in pleural fluid. *Chest* 2009; 136: 671–677.
30 Valdés L, José ES, Pose A, *et al.* Diagnostic value of N-terminal pro-brain natriuretic peptide in pleural effusions of cardiac origin. *Arch Bronconeumol* 2011; 47: 246–251.
31 Marinho FCA, Vargas FS, Fabri J, *et al.* Clinical usefulness of B-type natriuretic peptide in the diagnosis of pleural effusions due to heart failure. *Respirology* 2011; 16: 495–499.
32 Han Z-J, Wu X-D, Cheng J-J, *et al.* Diagnostic accuracy of natriuretic peptides for heart failure in patients with pleural effusion: a systematic review and updated meta-analysis. *PLoS One* 2015; 10: e0134376.
33 Porcel JM, Vives M, Cao G, *et al.* Measurement of pro-brain natriuretic peptide in pleural fluid for the diagnosis of pleural effusions due to heart failure. *Am J Med* 2004; 116: 417–420.
34 Porcel JM, Chorda J, Cao G, *et al.* Comparing serum and pleural fluid pro-brain natriuretic peptide (NT-proBNP) levels with pleural-to-serum albumin gradient for the identification of cardiac effusions misclassified by Light's criteria. *Respirology* 2007; 12: 654–659.
35 Porcel JM, Bielsa S, Morales-Rull JL, *et al.* Comparison of pleural N-terminal pro-B-type natriuretic peptide, mid-region pro-atrial natriuretic peptide and mid-region pro-adrenomedullin for the diagnosis of pleural effusions associated with cardiac failure. *Respirology* 2013; 18: 540–545.
36 Porcel JM, Ferreiro L, Civit C, *et al.* Development and validation of a scoring system for the identification of pleural exudates of cardiac origin. *Eur J Intern Med* 2018; 50: 60–64.
37 Jiménez D, Díaz G, Gil D, *et al.* Etiology and prognostic significance of massive pleural effusions. *Respir Med* 2005; 99: 1183–1187.
38 Maher GG, Berger HW. Massive pleural effusion: malignant and nonmalignant causes in 46 patients. *Am Rev Respir Dis* 1972; 105: 458–460.
39 Nandalur KR, Hardie AH, Bollampally SR, *et al.* Accuracy of computed tomography attenuation values in the characterization of pleural fluid: an ROC study. *Acad Radiol* 2005; 12: 987–991.
40 Abramowitz Y, Simanovsky N, Goldstein MS, *et al.* Pleural effusion: characterization with CT attenuation values and CT appearance. *AJR Am J Roentgenol* 2009; 192: 618–623.
41 Çullu N, Kalemci S, Karakaş Ö, *et al.* Efficacy of CT in diagnosis of transudates and exudates in patients with pleural effusion. *Diagn Interv Radiol* 2014; 20: 116–120.
42 Rashid R, Jalili J, Arhami S, *et al.* The accuracy of chest computed tomography findings in differentiation of exudative from transudative pleural effusion. *Ann Clin Anal Med* 2015; 6: 341–344.
43 Yalçin-Şafak K, Umarusman-Tanju N, Ayyıldız M, *et al.* Efficacy of computed tomography (CT) attenuation values and CT findings in the differentiation of pleural effusion. *Pol J Radiol* 2017; 82: 100–105.
44 Porcel JM, Baguìeste G, Pardina M, *et al.* Computed tomography attenuation values of pleural fluid are useless for differentiating transudates from exudates. *Spanish J Med* 2022; 2: 1.
45 Porcel JM, Pardina M, Bielsa S, *et al.* Derivation and validation of a CT scan scoring system for discriminating malignant from benign pleural effusions. *Chest* 2015; 147: 513–519.
46 Reuter S, Lindgaard D, Laursen C, *et al.* Computed tomography of the chest in unilateral pleural effusions: outcome of the British Thoracic Society guideline. *J Thorac Dis* 2019; 11: 1336–1346.
47 Chen H-J, Tu C-Y, Ling S-J, *et al.* Sonographic appearances in transudative pleural effusions: not always an anechoic pattern. *Ultrasound Med Biol* 2008; 34: 362–369.
48 Asciak R, Hassan M, Mercer RM, *et al.* Prospective analysis of the predictive value of sonographic pleural fluid echogenicity for the diagnosis of exudative effusion. *Respiration* 2019; 97: 451–456.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

49 Shkolnik B, Judson MA, Austin A, *et al.* Diagnostic accuracy of thoracic ultrasonography to differentiate transudative from exudative pleural effusion. *Chest* 2020; 158: 692–697.
50 Soni NJ, Dreyfuss ZS, Ali S, *et al.* Pleural fluid echogenicity measured by ultrasound image pixel density to differentiate transudative vs. exudative pleural effusions. *Ann Am Thorac Soc* 2022; 19: 857–860.
51 Wang T, Du G, Fang L, *et al.* Value of ultrasonography in determining the nature of pleural effusion: analysis of 582 cases. *Medicine* 2022; 101: e30119.
52 Shiroshita A, Nozaki S, Tanaka Y, *et al.* Thoracic ultrasound for malignant pleural effusion: a systematic review and meta-analysis. *ERJ Open Research* 2020; 6: 00464-2020.
53 Porcel JM. Pleural mesothelioma. *Med Clin (Barc)* 2022; 159: 240–247.
54 Tsakok M, Hallifax R. Updates in pleural imaging. *Clin Chest Med* 2021; 42: 577–590.
55 Inan N, Arslan A, Akansel G, *et al.* Diffusion-weighted MRI in the characterization of pleural effusions. *Diagn Interv Radiol* 2009; 15: 13–18.
56 Walker SP, Bintcliffe O, Keenan E, *et al.* Randomised trial of indwelling pleural catheters for refractory transudative pleural effusions. *Eur Respir J* 2022; 59: 2101362.
57 Krishnan M, Cheriyath P, Wert Y, *et al.* The untapped potential of tunneled pleural catheters. *Ann Thorac Surg* 2015; 100: 2055–2057.
58 Harris K, Chalhoub M. The use of a PleurX catheter in the management of recurrent benign pleural effusion: a concise review. *Heart Lung Circ* 2012; 21: 661–665.
59 Chalhoub M, Harris K, Castellano M, *et al.* The use of the PleurX catheter in the management of non-malignant pleural effusions. *Chron Respir Dis* 2011; 8: 185–191.
60 Patil M, Dhillon SS, Attwood K, *et al.* Management of benign pleural effusions using indwelling pleural catheters: a systematic review and meta-analysis. *Chest* 2017; 151: 626–635.
61 Frost N, Ruwwe-Glösenkamp C, Raspe M, *et al.* Indwelling pleural catheters for non-malignant pleural effusions: report on a single centre's 10 years of experience. *BMJ Open Respir Res* 2020; 7: e000501.
62 Murthy SC, Okereke I, Mason DP, *et al.* A simple solution for complicated pleural effusions. *J Thorac Oncol* 2006; 1: 697–700.
63 Aboudara M, Maldonado F. Indwelling pleural catheters for benign pleural effusions: what is the evidence? *Curr Opin Pulm Med* 2019; 25: 369–373.
64 Majid A, Kheir F, Fashjian M, *et al.* Tunneled pleural catheter placement with and without talc poudrage for treatment of pleural effusions due to congestive heart failure. *Ann Am Thorac Soc* 2016; 13: 212–216.
65 Bhatnagar R, Reid ED, Corcoran JP, *et al.* Indwelling pleural catheters for non-malignant effusions: a multicentre review of practice. *Thorax* 2014; 69: 959–961.
66 Freeman RK, Ascioti AJ, Dake M, *et al.* A propensity-matched comparison of pleurodesis or tunneled pleural catheter for heart failure patients with recurrent pleural effusion. *Ann Thorac Surg* 2014; 97: 1872–1876.
67 Srour N, Potechin R, Amjadi K. Use of indwelling pleural catheters for cardiogenic pleural effusions. *Chest* 2013; 144: 1603–1608.
68 Herlihy JP, Loyalka P, Gnananandh J, *et al.* PleurX catheter for the management of refractory pleural effusions in congestive heart failure. *Tex Heart Inst J* 2009; 36: 38–43.
69 Li P, Hosseini S, Zhang T, *et al.* Clinical predictors of successful and earlier removal of indwelling pleural catheters in benign pleural effusions. *Respiration* 2019; 98: 239–245.
70 Lazarevic A, Dobric M, Goronja B, *et al.* Lung ultrasound-guided therapeutic thoracentesis in refractory congestive heart failure. *Acta Cardiol* 2020; 75: 398–405.
71 Ekpe EE, Essien IO, Idongesit U. Significant pleural effusion in congestive heart failure necessitating pleural drainage. *Nigerian J Cardiol* 2015; 12: 106–110.
72 Sundaralingam A, Bedawi EO, Harriss EK, *et al.* The frequency, risk factors, and management of complications from pleural procedures. *Chest* 2022; 161: 1407–1425.
73 Wahidi MM, Reddy C, Yarmus L, *et al.* Randomized trial of pleural fluid drainage frequency in patients with malignant pleural effusions. The ASAP trial. *Am J Respir Crit Care Med* 2017; 195: 1050–1057.
74 Muruganandan S, Azzopardi M, Fitzgerald DB, *et al.* Aggressive *versus* symptom-guided drainage of malignant pleural effusion *via* indwelling pleural catheters (AMPLE-2): an open-label randomised trial. *Lancet Respir Med* 2018; 6: 671–680.
75 Maskell N, Lee Y, Gleeson F, *et al.* Randomized trials describing lung inflammation after pleurodesis with talc of varying particle size. *Am J Respir Crit Care Med* 2004; 170: 377–382.
76 Steger V, Mika U, Toomes H, *et al.* Who gains most? A 10-year experience with 611 thoracoscopic talc pleurodeses. *Ann Thorac Surg* 2007; 83: 1940–1945.
77 Glazer M, Berkman N, Lafair JS, *et al.* Successful talc slurry pleurodesis in patients with nonmalignant pleural effusion. *Chest* 2000; 117: 1404–1409.
78 Little AG, Kadowaki MH, Ferguson MK, *et al.* Pleuro-peritoneal shunting. Alternative therapy for pleural effusions. *Ann Surg* 1988; 208: 443–450.
79 Artemiou O, Marta G-M, Klepetko W, *et al.* Pleurovenous shunting in the treatment of nonmalignant pleural effusion. *Ann Thorac Surg* 2003; 76: 231–233.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

80 Morales-Rull JL, Bielsa S, Conde-Martel A, *et al.* Pleural effusions in acute decompensated heart failure: prevalence and prognostic implications. *Eur J Intern Med* 2018; 52: 49–53.

81 Muruganandan S, Azzopardi M, Thomas R, *et al.* The Pleural Effusion And Symptom Evaluation (PLEASE) study of breathlessness in patients with a symptomatic pleural effusion. *Eur Respir J* 2020; 55: 1900980.

82 McDonagh TA, Metra M, Adamo M, *et al.* 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021; 42: 3599–3726.

83 Duerden L, Benamore R, Edey A. Radiology: what is the role of chest radiographs, CT and PET in modern management? *In*: Maskell NA, Laursen CB, Lee YCG, eds. Pleural Disease (ERS Monograph). Sheffield, European Respiratory Society, 2020; pp. 48–72.

84 Banka R, Skaarup S, Mercer R, *et al.* Thoracic ultrasound: a key tool beyond procedure guidance. *In*: Maskell NA, Laursen CB, Lee YCG, eds. Pleural Disease (ERS Monograph). Sheffield, European Respiratory Society, 2020; pp. 73–89.

85 Rosmini S, Seraphim A, Knott K, *et al.* Non-invasive characterization of pleural and pericardial effusions using T1 mapping by magnetic resonance imaging. *Eur Heart J Cardiovasc Imaging* 2022; 23: 1117–1126.

86 Li H, Chen J, Hu P-X. Cardiopulmonary ultrasound correlates of pleural effusions in patients with congestive heart failure. *BMC Cardiovasc Disord* 2022; 22: 198.

87 Via G, Hussain A, Wells M, *et al.* International evidence-based recommendations for focused cardiac ultrasound. *J Am Soc Echocardiogr* 2014; 27: e1–e33.

88 Maw AM, Hassanin A, Ho PM, *et al.* Diagnostic accuracy of point-of-care lung ultrasonography and chest radiography in adults with symptoms suggestive of acute decompensated heart failure: a systematic review and meta-analysis. *JAMA Netw Open* 2019; 2: e190703.

89 Miger KC, Fabricius-Bjerre A, Maschmann CP, *et al.* Clinical applicability of lung ultrasound methods in the emergency department to detect pulmonary congestion on computed tomography. *Ultraschall Med* 2021; 42: e21–e30.

90 Kataoka H, Takada S. The role of thoracic ultrasonography for evaluation of patients with decompensated chronic heart failure. *J Am Coll Cardiol* 2000; 35: 1638–1646.

91 Gallard E, Redonnet J-P, Bourcier J-E, *et al.* Diagnostic performance of cardiopulmonary ultrasound performed by the emergency physician in the management of acute dyspnea. *Am J Emerg Med* 2015; 33: 352–358.

92 Pivetta E, Goffi A, Lupia E, *et al.* Lung ultrasound-implemented diagnosis of acute decompensated heart failure in the ED: a SIMEU multicenter study. *Chest* 2015; 148: 202–210.

93 Arvig MD, Laursen CB, Jacobsen N, *et al.* Monitoring patients with acute dyspnea with serial point-of-care ultrasound of the inferior vena cava (IVC) and the lungs (LUS): a systematic review. *J Ultrasound* 2022; 25: 547–561.

94 Anderson KL, Jenq KY, Fields JM, *et al.* Diagnosing heart failure among acutely dyspneic patients with cardiac, inferior vena cava, and lung ultrasonography. *Am J Emerg Med* 2013; 31: 1208–1214.

95 Laursen CB, Clive A, Hallifax R, *et al.* European Respiratory Society statement on thoracic ultrasound. *Eur Respir J* 2021; 57: 2001519.

96 Qureshi NR, Rahman NM, Gleeson FV. Thoracic ultrasound in the diagnosis of malignant pleural effusion. *Thorax* 2009; 64: 139–143.

97 Falster C, Jacobsen N, Coman KE, *et al.* Diagnostic accuracy of focused deep venous, lung, cardiac and multiorgan ultrasound in suspected pulmonary embolism: a systematic review and meta-analysis. *Thorax* 2022; 77: 679–689.

98 Bedawi EO, Kanellakis NI, Corcoran JP, *et al.* The biological role of pleural fluid PAI-1 and sonographic septations in pleural infection: analysis of a prospectively collected clinical outcome study. *Am J Respir Crit Care Med* 207: 731–739.

99 Torres-Macho J, Cerqueiro-González JM, Arévalo-Lorido JC, *et al.* The effects of a therapeutic strategy guided by lung ultrasound on 6-month outcomes in patients with heart failure: results from the EPICC randomized controlled trial. *J Clin Med* 2022; 11: 4930.

100 Araiza-Garaygordobil D, Gopar-Nieto R, Martinez-Amezcua P, *et al.* A randomized controlled trial of lung ultrasound-guided therapy in heart failure (CLUSTER-HF study). *Am Heart J* 2020; 227: 31–39.

101 Porcel JM, Mas E, Rene JM, *et al.* Hidrotórax de causa hepática: análisis de 77 pacientes [Hepatic hydrothorax: report of a series of 77 patients]. *Med Clin (Barc)* 2013; 141: 484–486.

102 Singh A, Bajwa A, Shujaat A. Evidence-based review of the management of hepatic hydrothorax. *Respiration* 2013; 86: 155–173.

103 Hung T-H, Tseng C-W, Tsai C-C, *et al.* Mortality following catheter drainage *versus* thoracentesis in cirrhotic patients with pleural effusion. *Dig Dis Sci* 2017; 62: 1080–1085.

104 O'Leary JG, Rajender Reddy K, Tandon P, *et al.* Increased risk of ACLF and inpatient mortality in hospitalized patients with cirrhosis and hepatic hydrothorax. *Dig Dis Sci* 2021; 66: 3612–3618.

105 Castellote J, Xiol X, Cortés-Beut R, *et al.* Complications of thoracentesis in cirrhotic patients with pleural effusion. *Rev Esp Enferm Dig* 2001; 93: 566–575.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

106 Shojaee S, Khalid M, Kallingal G, *et al.* Repeat thoracentesis in hepatic hydrothorax and non-hepatic hydrothorax effusions: a case-control study. *Respiration* 2018; 96: 330–337.
107 Romero S, Lim AK, Singh G, *et al.* Natural history and outcomes of patients with liver cirrhosis complicated by hepatic hydrothorax. *World J Gastroenterol* 2022; 28: 5175–5187.
108 Han S-K, Kang S-H, Kim M-Y, *et al.* Outcome of intermittent thoracentesis *versus* pigtail catheter drainage for hepatic hydrothorax. *J Clin Med* 2022; 11: 7221.
109 Adlakha N, Russo MW. Outcomes after transjugular intrahepatic portosystemic shunt in cirrhotic patients 70 years and older. *J Clin Med* 2020; 9: 381.
110 Badillo R, Rockey DC. Hepatic hydrothorax: clinical features, management, and outcomes in 77 patients and review of the literature. *Medicine (Baltimore)* 2014; 93: 135–142.
111 Bisht RU, Liu MC, Koblinski JE, *et al.* Is 70 the new 50? Complications and outcomes of transjugular intrahepatic portosystemic shunt in older *versus* younger patients. *Abdom Radiol (NY)* 2021; 46: 2789–2794.
112 Dhanasekaran R, West JK, Gonzales PC, *et al.* Transjugular intrahepatic portosystemic shunt for symptomatic refractory hepatic hydrothorax in patients with cirrhosis. *Am J Gastroenterol* 2010; 105: 635–641.
113 Jindal A, Mukund A, Kumar G, *et al.* Efficacy and safety of transjugular intrahepatic portosystemic shunt in difficult-to-manage hydrothorax in cirrhosis. *Liver Int* 2019; 39: 2164–2173.
114 Young S, Bermudez J, Zhang L, *et al.* Transjugular intrahepatic portosystemic shunt (TIPS) placement: a comparison of outcomes between patients with hepatic hydrothorax and patients with refractory ascites. *Diagn Interv Imaging* 2019; 100: 303–308.
115 Deltenre P, Zanetto A, Saltini D, *et al.* The role of transjugular intrahepatic portosystemic shunt in patients with cirrhosis and ascites: recent evolution and open questions. *Hepatology* 2023; 77: 640–658.
116 Mahmoud EM, Anwar MIA. Comparison of tigecycline and bleomycin pleurodesis by pigtail catheter in hepatic hydrothorax with liver cirrhosis. *Egypt J Chest Dis Tuberc* 2020; 69: 98.
117 Lee WJ, Kim HJ, Park JH, *et al.* Chemical pleurodesis for the management of refractory hepatic hydrothorax in patients with decompensated liver cirrhosis. *Korean J Hepatol* 2011; 17: 292–298.
118 Chen A, Massoni J, Jung D, *et al.* Indwelling tunneled pleural catheters for the management of hepatic hydrothorax. A pilot study. *Ann Am Thorac Soc* 2016; 13: 862–866.
119 Kniese C, Diab K, Ghabril M, *et al.* Indwelling pleural catheters in hepatic hydrothorax: a single-center series of outcomes and complications. *Chest* 2019; 155: 307–314.
120 Shojaee S, Rahman N, Haas K, *et al.* Indwelling tunneled pleural catheters for refractory hepatic hydrothorax in patients with cirrhosis: a multicenter study. *Chest* 2019; 155: 546–553.
121 Alhabeeb FF, Carle-Talbot K, Rakocevic N, *et al.* Indwelling tunneled pleural catheters in patients with hepatic hydrothorax: a single-center analysis for outcomes and complications. *Can J Respir Crit Care Sleep Med* 2023; 7: 4–9.
122 Assouad J, Barthes FLP, Shaker W, *et al.* Recurrent pleural effusion complicating liver cirrhosis. *Ann Thorac Surg* 2003; 75: 986–989.
123 Luh S-P, Chen C-Y. Video-assisted thoracoscopic surgery (VATS) for the treatment of hepatic hydrothorax: report of twelve cases. *J Zhejiang Univ Sci B* 2009; 10: 547–551.
124 Milanez de Campos JR, Filho LO, de Campos Werebe E, *et al.* Thoracoscopy and talc poudrage in the management of hepatic hydrothorax. *Chest* 2000; 118: 13–17.
125 Xiol X, Tremosa G, Castellote J, *et al.* Liver transplantation in patients with hepatic hydrothorax. *Transpl Int* 2005; 18: 672–675.
126 Jarratt MJ, Sahn SA. Pleural effusions in hospitalized patients receiving long-term hemodialysis. *Chest* 1995; 108: 470–474.
127 Bakirci T, Sasak G, Ozturk S, *et al.* Pleural effusion in long-term hemodialysis patients. *Transplant Proc* 2007; 39: 889–891.
128 Lakadamyali H, Lakadamyali H, Ergun T. Thorax CT findings in symptomatic hemodialysis patients. *Transplant Proc* 2008; 40: 71–76.
129 Coşkun M, Boyvat F, Bozkurt B, *et al.* Thoracic CT findings in long-term hemodialysis patients. *Acta Radiol* 1999; 40: 181–186.
130 Qureshi SQ, Idrees MK, Ahmad S, *et al.* Pleural effusion among patients on maintenance hemodialysis at SIUT Karachi, Pakistan. *Rawal Med J* 2016; 41: 11.
131 Hamada S, Sano T, Nagatani Y, *et al.* Pleural effusion negatively impacts survival of patients undergoing maintenance hemodialysis. *Pulmonology* 2019; 25: 58–60.
132 Shaik L, Thotamgari SR, Kowtha P, *et al.* A spectrum of pulmonary complications occurring in end-stage renal disease patients on maintenance hemodialysis. *Cureus* 2021; 13: e15426.
133 Jabbar A, Qureshi R, Nasir K, *et al.* Transudative and exudative pleural effusion in chronic kidney disease patients: a prospective single-center study. *Cureus* 2021; 13: e18649.
134 Ray S, Mukherjee S, Ganguly J, *et al.* A cross-sectional prospective study of pleural effusion among cases of chronic kidney disease. *Indian J Chest Dis Allied Sci* 2013; 55: 209–213.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

135 Uzan G, İkitimur H. Pleural effusion in end stage renal failure patients. *Sisli Etfal Hastan Tip Bul* 2019; 53: 54–57.
136 Pant P, Baniya S, Jha A. Prevalence of respiratory manifestations in chronic kidney diseases; a descriptive cross-sectional study in a tertiary care hospital of Nepal. *JNMA J Nepal Med Assoc* 2019; 57: 80–83.
137 Wu J, Lin L, Jiang X, *et al.* Characteristics and negative impacts of pleural effusion in hospitalized patients undergoing maintenance hemodialysis. *Am J Transl Res* 2022; 14: 7494–7503.
138 Berger HW, Rammohan G, Neff MS, *et al.* Uremic pleural effusion. A study in 14 patients on chronic dialysis. *Ann Intern Med* 1975; 82: 362–364.
139 Maher JF. Uremic pleuritis. *Am J Kidney Dis* 1987; 10: 19–22.
140 Karki A, Riley L, Mehta HJ, *et al.* Pleural effusions of urinary etiologies. *Dis Mon* 2019; 65: 104–108.
141 Spithoven EM, Abrahams AC. Unilateral pleural effusion in a hemodialysis patient. *Kidney360* 2021; 2: 1542–1543.
142 Shemin D, Clark DD, Chazan JA. Unexplained pleural effusions in the peritoneal dialysis population. *Perit Dial Int* 1989; 9: 143.
143 Doelken P, Huggins JT, Goldblatt M, *et al*. Effects of coexisting pneumonia and end-stage renal disease on pleural fluid analysis in patients with hydrostatic pleural effusion. *Chest* 2013; 143: 1709–1716.
144 Corbett RW, Ashby DR. Complication of diagnostic pleural aspiration: is it of value in hemodialysis patients? *Hemodial Int* 2014; 18: 546–550.
145 Kwan BC-H, Chow K-M, Pang W-F, *et al.* Unexplained exudative pleural effusion in chronic peritoneal dialysis patients. *Perit Dial Int* 2010; 30: 534–540.
146 Goodkin DA, Bragg-Gresham JL, Koenig KG, *et al.* Association of comorbid conditions and mortality in hemodialysis patients in Europe, Japan, and the United States: the Dialysis Outcomes and Practice Patterns Study (DOPPS). *J Am Soc Nephrol* 2003; 14: 3270–3277.
147 DeBiasi EM, Pisani MA, Murphy TE, *et al.* Mortality among patients with pleural effusion undergoing thoracentesis. *Eur Respir J* 2015; 46: 495–502.
148 Potechin R, Amjadi K, Srour N. Indwelling pleural catheters for pleural effusions associated with end-stage renal disease: a case series. *Ther Adv Respir Dis* 2015; 9: 22–27.
149 Colella S, Fioretti F, Massaccesi C, *et al.* Usefulness of medical thoracoscopy in the management of pleural effusion caused by chronic renal failure. *J Bronchology Interv Pulmonol* 2017; 24: 285–289.
150 Horita Y, Noguchi M, Miyazaki M, *et al.* Prognosis of patients with rounded atelectasis undergoing long-term hemodialysis. *NEF Karger Publishers* 2001; 88: 87–92.
151 Gilbert L, Ribot S, Frankel H, *et al.* Fibrinous uremic pleuritis: a surgical entity. *Chest* 1975; 67: 53–56.
152 Rodelas R, Rakowski TA, Argy WP, *et al.* Fibrosing uremic pleuritis during hemodialysis. *JAMA* 1980; 243: 2424–2425.
153 Chow KM, Szeto CC, Li PK-T. Management options for hydrothorax complicating peritoneal dialysis. *Semin Dial* 2003; 16: 389–394.
154 Nomoto Y, Suga T, Nakajima K, *et al.* Acute hydrothorax in continuous ambulatory peritoneal dialysis – a collaborative study of 161 centers. *Am J Nephrol* 1989; 9: 363–367.
155 Ramon RG, Carrasco AM. Hydrothorax in peritoneal dialysis. *Perit Dial Int* 1998; 18: 5–10.
156 Lorch DG, Sahn SA. Pleural effusions due to diseases below the diaphragm. *Semin Respir Med* 1987; 9: 75–85.
157 Chow KM, Szeto CC, Wong TY-H, *et al.* Hydrothorax complicating peritoneal dialysis: diagnostic value of glucose concentration in pleural fluid aspirate. *Perit Dial Int* 2002; 22: 525–527.
158 Momenin N, Colletti PM, Kaptein EM. Low pleural fluid-to-serum glucose gradient indicates pleuroperitoneal communication in peritoneal dialysis patients: presentation of two cases and a review of the literature. *Nephrol Dial Transplant* 2012; 27: 1212–1219.
159 Harry L, Nyakale N, Tinarwo P. Scintigraphic peritoneography in the diagnosis of pleuroperitoneal leak complicating peritoneal dialysis: a comparison with conventional diagnostic methods. *Medicine* 2020; 99: e21029.
160 Matsuoka N, Yamaguchi M, Asai A, *et al.* The effectiveness and safety of computed tomographic peritoneography and video-assisted thoracic surgery for hydrothorax in peritoneal dialysis patients: a retrospective cohort study in Japan. *PLoS One* 2020; 15: e0238602.
161 Inanaga R, Oda M, Asahina K, *et al.* The new method to make diagnosis and identify the location of leakage of pleuroperitoneal communication in peritoneal dialysis patients. *CEN Case Rep* 2022; 11: 471–476.
162 Prasad N, Patel MR, Kushwaha R, *et al.* Modalities of diagnosis and management of peritoneal dialysis-related hydrothorax including videothoracoscopy-assisted repair: a single-center experience. *Indian J Nephrol* 2021; 31: 574.
163 Nemeth A, Mustafi M, Friedel G, *et al.* Thoracoscopic mesh implantation as a definitive treatment approach for peritoneal dialysis-associated hydrothorax. *Updates Surg* 2022; 74: 2011–2017.
164 Chen H-YM, Chan H-YH, Chan H-MH, *et al.* Surgical management of pleuro-peritoneal fistula in chronic renal failure patient—safety and effectiveness. *J Thorac Dis* 2021; 13: 2979–2985.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

165 Epler GR, McLoud TC, Gaensler EA. Prevalence and incidence of benign asbestos pleural effusion in a working population. *JAMA* 1982; 247: 617–622.
166 Mastrangelo G, Ballarin MN, Bellini E, *et al.* Asbestos exposure and benign asbestos diseases in 772 formerly exposed workers: dose-response relationships. *Am J Ind Med* 2009; 52: 596–602.
167 Fujimoto N, Gemba K, Aoe K, *et al.* Clinical investigation of benign asbestos pleural effusion. *Pulm Med* 2015; 2015: 1–6.
168 Fujimoto N, Kato K, Usami I, *et al.* Asbestos-related diffuse pleural thickening. *Respiration* 2014; 88: 277–284.
169 Schwartz DA, Galvin JR, Dayton CS, *et al.* Determinants of restrictive lung function in asbestos-induced pleural fibrosis. *J Appl Physiol (1985)* 1990; 68: 1932–1937.
170 Guidotti TL, Miller A, Christiani D, *et al.* Diagnosis and initial management of nonmalignant diseases related to asbestos. *Am J Respir Crit Care Med* 2004; 170: 691–715.
171 Nojima D, Fujimoto N, Kato K, *et al.* Pilot analysis of asbestos-induced diffuse pleural thickening with respiratory compromise. *Acta Med Okayama* 2015; 69: 261–266.
172 Hara R, Yano Y, Okabe F, *et al.* Radiographic change over 11 years in a patient with asbestos-related pleural disease. *Respirol Case Rep* 2020; 8: e00642.
173 Hillerdal G, Ozesmi M. Benign asbestos pleural effusion: 73 exudates in 60 patients. *Eur J Respir Dis* 1987; 71: 113–121.
174 Jeebun V, Stenton SC. The presentation and natural history of asbestos-induced diffuse pleural thickening. *Occup Med* 2012; 62: 266–268.
175 Robinson BW, Musk AW. Benign asbestos pleural effusion: diagnosis and course. *Thorax* 1981; 36: 896–900.
176 Stephens M, Gibbs AR, Pooley FD, *et al.* Asbestos induced diffuse pleural fibrosis: pathology and mineralogy. *Thorax* 1987; 42: 583–588.
177 Allen RKA, Cramond T, Lennon D, *et al.* A retrospective study of chest pain in benign asbestos pleural disease. *Pain Med* 2011; 12: 1303–1308.
178 Luo W, Zeng Y, Shen P, *et al.* A multidisciplinary approach for the diagnosis of benign asbestos pleural effusion: a single-center experience. *J Thorac Dis* 2020; 12: 4338–4346.
179 Kishimoto T, Kato K, Ashizawa K, *et al.* A retrospective study on radiological findings of diffuse pleural thickening with benign asbestos pleural effusion in Japanese cases. *Ind Health* 2022; 60: 429–435.
180 Fonseka DD, Edey A, Stadon L, *et al.* The physiological consequences of different distributions of diffuse pleural thickening on CT imaging. *Br J Radiol* 2017; 90: 20170218.
181 Metintas M, Ucgun I, Elbek O, *et al.* Computed tomography features in malignant pleural mesothelioma and other commonly seen pleural diseases. *Eur J Radiol* 2002; 41: 1–9.
182 Kato K, Gemba K, Fujimoto N, *et al.* Pleural irregularities and mediastinal pleural involvement in early stages of malignant pleural mesothelioma and benign asbestos pleural effusion. *Eur J Radiol* 2016; 85: 1594–1600.
183 Leung AN, Müller NL, Miller RR. CT in differential diagnosis of diffuse pleural disease. *AJR Am J Roentgenol* 1990; 154: 487–492.
184 Okten F, Köksal D, Onal M, *et al.* Computed tomography findings in 66 patients with malignant pleural mesothelioma due to environmental exposure to asbestos. *Clin Imaging* 2006; 30: 177–180.
185 Terada T, Tabata C, Tabata R, *et al.* Clinical utility of 18-fluorodeoxyglucose positron emission tomography/computed tomography in malignant pleural mesothelioma. *Exp Ther Med* 2012; 4: 197–200.
186 Khan AM, Tlemcani K, Shanmugam N YD, *et al.* A localized pleural based mass with intense uptake on positron emission tomography scan. *Chest* 2007; 131: 294–299.
187 Yildirim H, Metintas M, Entok E, *et al.* Clinical value of fluorodeoxyglucose-positron emission tomography/computed tomography in differentiation of malignant mesothelioma from asbestos-related benign pleural disease: an observational pilot study. *J Thorac Oncol* 2009; 4: 1480–1484.
188 International Labour Organization. Guidelines for the use of the ILO International Classification of Radiographs of Pneumoconioses - revised edition 2022. www.ilo.org/global/topics/safety-and-health-at-work/resources-library/publications/WCMS_867859/lang--en/index.htm Date last accessed: 3 May 2023.
189 Lynch DA, Gamsu G, Aberle DR. Conventional and high resolution computed tomography in the diagnosis of asbestos-related diseases. *Radiographics* 1989; 9: 523–551.
190 Gevenois PA, de Maertelaer V, Madani A, *et al.* Asbestosis, pleural plaques and diffuse pleural thickening: three distinct benign responses to asbestos exposure. *Eur Respir J* 1998; 11: 1021–1027.
191 Schaal M, Severac F, Labani A, *et al.* Diagnostic performance of ultra-low-dose computed tomography for detecting asbestos-related pleuropulmonary diseases: prospective study in a screening setting. *PLoS One* 2016; 11: e0168979.
192 Kramer H, Pieterman RM, Slebos D-J, *et al.* PET for the evaluation of pleural thickening observed on CT. *J Nucl Med* 2004; 45: 995–998.
193 Alì G, Bruno R, Fontanini G. The pathological and molecular diagnosis of malignant pleural mesothelioma: a literature review. *J Thorac Dis* 2018; 10: S276–S284.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

194 Hoyle JL, Gudur S. P206 Asbestos related diffuse pleural thickening; Likelihood of progression in the secondary care setting. *Thorax* 2013; 68: Suppl. 3, A169.

195 Ferguson K, Mercer R, King J, *et al.* S43 Preliminary results of the Meso-ORIGINS feasibility study: retrospective element regarding BAPE-mesothelioma evolution rate. *Thorax* 2021; 76: Suppl. 1, A28.

196 Ferguson KJ, Blyth KG, Neilson M. S15 Evolution of mesothelioma following initial biopsies showing benign pleural inflammation: a meta-analysis. *Thorax* 2021; 76: Suppl. 2, A14.

197 Meso-ORIGINS: Mesothelioma Observational study of RIsk prediction and Generation of benign-meso tissue pairs, Including a Nested MRI Sub-study. www.hra.nhs.uk/planning-and-improving-research/application-summaries/research-summaries/meso-origins/ Date last accessed: 3 May 2023.

198 al Jarad N, Poulakis N, Pearson MC, *et al.* Assessment of asbestos-induced pleural disease by computed tomography – correlation with chest radiograph and lung function. *Respir Med* 1991; 85: 203–208.

199 Kee ST, Gamsu G, Blanc P. Causes of pulmonary impairment in asbestos-exposed individuals with diffuse pleural thickening. *Am J Respir Crit Care Med* 1996; 154: 789–793.

200 Brims F, Musk AW. Asbestos-related pleural diseases. *In*: Light RW, Lee YCG, eds. Textbook of Pleural Diseases. 3rd Edn. Boca Raton, CRC Press, 2016; pp. 464–472.

201 Fischer M-O, Brotons F, Briant AR, *et al.* Postoperative pulmonary complications after cardiac surgery: the VENICE international cohort study. *J Cardiothoracic Vasc Anesth* 2022; 36: 2344–2351.

202 Savas Bozbas S, Ulubay G, Sarinc Ulasli S, *et al.* Risk factors for post operative pulmonary complications and mortality in cardiac transplantation patients. *Ann Transplant* 2009; 14: 33–39.

203 Labidi M, Baillot R, Dionne B, *et al.* Pleural effusions following cardiac surgery: prevalence, risk factors, and clinical features. *Chest* 2009; 136: 1604–1611.

204 Light RW, Rogers JT, Moyers JP, *et al.* Prevalence and clinical course of pleural effusions at 30 days after coronary artery and cardiac surgery. *Am J Respir Crit Care Med* 2002; 166: 1567–1571.

205 Jain A, Devarajan A, Assallum H, *et al.* Characteristics of early pleural effusions after orthotopic heart transplantation: comparison with coronary artery bypass graft surgery. *Pleura Peritoneum* 2021; 6: 161–165.

206 Schiefenhövel F, Poncette A-S, Boyle EM, *et al.* Pleural effusions are associated with adverse outcomes after cardiac surgery: a propensity-matched analysis. *J Cardiothoracic Surg* 2022; 17: 298.

207 Miserocchi G, Berett E. Pathophysiology of Extravascular Water in the Pleural Cavity and in the Lung Interstitium After Lung Thoracic Surgery. *In*: Cardoso P, ed. Topics in Thoracic Surgery. London, IntchOpen, 2012. www.intechopen.com/books/topics-in-thoracic-surgery/pathophysiology-of-extravascular-water-in-the-pleural-cavity-and-in-the-lung-interstitium-after-lung

208 Barker WL. Natural history of residual air spaces after pulmonary resection. *Chest Surg Clin N Am* 1996; 6: 585–613.

209 Usta E, Mustafi M, Ziemer G. Ultrasound estimation of volume of postoperative pleural effusion in cardiac surgery patients. *Interact Cardiovasc Thorac Surg* 2010; 10: 204–207.

210 Xing T, Li X, Liu J, *et al.* Early removal of chest tubes leads to better short-term outcome after video-assisted thoracoscopic surgery lung resection. *Ann Transl Med* 2020; 8: 101.

211 Hansen LS, Hjortdal VE, Jakobsen CJ, *et al.* Early, dedicated follow-up and treatment of pleural effusions enhance the recovery rate after open cardiac surgery: results from a randomized, clinical trial. *Eur J Cardiothoracic Surg* 2017; 51: 58–66.

212 Tamarappoo BK, Klein AL. Post-pericardiotomy syndrome. *Curr Cardiol Rep* 2016; 18: 116.

213 Adler Y, Charron P, Imazio M, *et al.* ESC guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J* 2015; 2015: 2921–2964.

214 Imazio M, Trinchero R, Brucato A, *et al.* COlchicine for the Prevention of the Post-pericardiotomy Syndrome (COPPS): a multicentre, randomized, double-blind, placebo-controlled trial. *Eur Heart J* 2010; 31: 2749–2754.

215 Bonacchi M, Prifti E, Giunti G, *et al.* Respiratory dysfunction after coronary artery bypass grafting employing bilateral internal mammary arteries: the influence of intact pleura. *Eur J Cardiothorac Surg* 2001; 19: 827–833.

216 Ad N, Suyderhoud JP, Kim YD, *et al.* Benefits of prophylactic continuous infusion of furosemide after the maze procedure for atrial fibrillation. *J Thorac Cardiovasc Surg* 2002; 123: 232–236.

217 Cerfolio RJ, Bryant AS. Results of a prospective algorithm to remove chest tubes after pulmonary resection with high output. *J Thorac Cardiovasc Surg* 2008; 135: 269–273.

218 Bjerregaard LS, Jensen K, Petersen RH, *et al.* Early chest tube removal after video-assisted thoracic surgery lobectomy with serous fluid production up to 500 mL/day. *Eur J Cardiothorac Surg* 2014; 45: 241–246.

219 Wahidi MM, Willner DA, Snyder LD, *et al.* Diagnosis and outcome of early pleural space infection following lung transplantation. *Chest* 2009; 135: 484–491.

220 Shitrit D, Izbicki G, Fink G, *et al.* Late postoperative pleural effusion following lung transplantation: characteristics and clinical implications. *Eur J Cardiothorac Surg* 2003; 23: 494–496.

221 Misra H, Dikensoy O, Rodriguez RM, *et al.* Prevalence of pleural effusions post orthotopic heart transplantation. *Respirology* 2007; 12: 887–890.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

222 Rahman NM, Ali N, Brown G, *et al.* Local anaesthetic thoracoscopy: British Thoracic Society pleural disease guideline 2010. *Thorax* 2010; 65: ii54–ii60.
223 Sundaralingam A, Aujayeb A, Jackson KA, *et al.* Investigation and outcomes in patients with nonspecific pleuritis: results from the International Collaborative Effusion database. *ERJ Open Res* 2023; 9: 00599-2022.
224 Boutin C, Viallat JR, Cargnino P, *et al.* Thoracoscopy in malignant pleural effusions. *Am Rev Respir Dis* 1981; 124: 588–592.
225 Page RD, Jeffrey RR, Donnelly RJ. Thoracoscopy: a review of 121 consecutive surgical procedures. *Ann Thorac Surg* 1989; 48: 66–68.
226 Hucker J, Bhatnagar NK, al-Jilaihawi AN, *et al.* Thoracoscopy in the diagnosis and management of recurrent pleural effusions. *Ann Thorac Surg* 1991; 52: 1145–1147.
227 Menzies R, Charbonneau M. Thoracoscopy for the diagnosis of pleural disease. *Ann Intern Med* 1991; 114: 271–276.
228 Ohri SK, Oswal SK, Townsend ER, *et al.* Early and late outcome after diagnostic thoracoscopy and talc pleurodesis. *Ann Thorac Surg* 1992; 53: 1038–1041.
229 Kendall SW, Bryan AJ, Large SR, *et al.* Pleural effusions: is thoracoscopy a reliable investigation? A retrospective review. *Respir Med* 1992; 86: 437–440.
230 Hansen M, Faurschou P, Clementsen P. Medical thoracoscopy, results and complications in 146 patients: a retrospective study. *Respir Med* 1998; 92: 228–232.
231 Blanc F-X, Atassi K, Bignon J, *et al.* Diagnostic value of medical thoracoscopy in pleural disease: a 6-year retrospective study. *Chest* 2002; 121: 1677–1683.
232 Janssen JP, Ramlal S, Mravunac M. The long-term follow up of exudative pleural effusion after nondiagnostic thoracoscopy. *J Bronchol Interv Pulmonol* 2004; 11: 169–174.
233 Davies HE, Nicholson JE, Rahman NM, *et al.* Outcome of patients with nonspecific pleuritis/fibrosis on thoracoscopic pleural biopsies. *Eur J Cardiothorac Surg* 2010; 38: 472–477.
234 Metintas M, Ak G, Cadirci O, *et al.* Outcome of patients diagnosed with fibrinous pleuritis after medical thoracoscopy. *Respir Med* 2012; 106: 1177–1183.
235 Vakil E, Ost D, Vial MR, *et al.* Non-specific pleuritis in patients with active malignancy. *Respirology* 2018; 23: 213–219.
236 Yu Y-X, Yang Y, Wu Y-B, *et al.* An update of the long-term outcome of patients with nonspecific pleurisy at medical thoracoscopy. *BMC Pulmonary Med* 2021; 21: 226.
237 Reuter SB, Clementsen PF, Bodtger U. Incidence of malignancy and survival in patients with idiopathic pleuritis. *J Thorac Dis* 2019; 11: 386–392.
238 Venekamp LN, Velkeniers B, Noppen M. Does 'idiopathic pleuritis' exist? Natural history of non-specific pleuritis diagnosed after thoracoscopy. *Respiration* 2005; 72: 74–78.
239 DePew ZS, Verma A, Wigle D, *et al.* Nonspecific pleuritis: optimal duration of follow-up. *Ann Thorac Surg* 2014; 97: 1867–1871.
240 Gunluoglu G, Olcmen A, Gunluoglu MZ, *et al.* Long-term outcome of patients with undiagnosed pleural effusion. *Arch Bronconeumol* 2015; 51: 632–636.
241 Ferrer JS, Muñoz XG, Orriols RM, *et al.* Evolution of idiopathic pleural effusion: a prospective, long-term follow-up study. *Chest* 1996; 109: 1508–1513.
242 Karpathiou G, Anevlavis S, Tiffet O, *et al.* Clinical long-term outcome of non-specific pleuritis (NSP) after surgical or medical thoracoscopy. *J Thorac Dis* 2020; 12: 2096–2104.
243 Churg A, Sheffield BS, Galateau-Salle F. New markers for separating benign from malignant mesothelial proliferations: are we there yet? *Arch Pathol Lab Med* 2016; 140: 318–321.
244 Brcic L, Le Stang N, Gallob F, *et al.* A combination of MTAP and p16 immunohistochemistry can substitute for CDKN2A fluorescence *in situ* hybridization in diagnosis and prognosis of pleural mesotheliomas. *Arch Pathol Lab Med* 2023; 147: 313–322.
245 Berg KB, Dacic S, Miller C, *et al.* Utility of methylthioadenosine phosphorylase compared with BAP1 immunohistochemistry, and CDKN2A and NF2 fluorescence *in situ* hybridization in separating reactive mesothelial proliferations from epithelioid malignant mesotheliomas. *Arch Pathol Lab Med* 2018; 142: 1549–1553.
246 Karpathiou G, Péoc'h M, Sundaralingam A, *et al.* Inflammation of the pleural cavity: a review on pathogenesis, diagnosis and implications in tumor pathophysiology. *Cancers (Basel)* 2022; 14: 1415.
247 Klebe S, Galateau Salle F, Bruno R, *et al.* The highlights of the 15th International Conference of the International Mesothelioma Interest Group - do concepts challenge the traditional approach to pathological mesothelioma diagnosis? *Lung Cancer* 2022; 163: 1–6.
248 Klebe S, Nakatani Y, Dobra K, *et al.* The concept of mesothelioma *in situ*, with consideration of its potential impact on cytology diagnosis. *Pathology* 2021; 53: 446–453.
249 Hashimoto M, Sato A, Kuroda A, *et al.* Clinical feature of diagnostic challenging cases for pleural biopsy in patient with malignant pleural mesothelioma. *Gen Thorac Cardiovasc Surg* 2020; 68: 820–827.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.

250 Jiang W, Han Z, Tang X, *et al.* Diffusion-weighted imaging diagnostic algorithm in patients with suspected pleural malignancy. *Eur Radiol* 2021; 31: 9038–9047.

251 Revelli M, Chiesa F, Del Prato A, *et al.* Role of respiratory-triggered diffusion-weighted MRI in the assessment of pleural disease. *Br J Radiol* 2016; 89: 20160289.

252 Porcel JM. Biomarkers in the diagnosis of pleural diseases: a 2018 update. *Ther Adv Respir Dis* 2018; 12: 1753466618808660.

253 Schillebeeckx E, van Meerbeeck JP, Lamote K. Clinical utility of diagnostic biomarkers in malignant pleural mesothelioma: a systematic review and meta-analysis. *Eur Respir Rev* 2021; 30: 210057.

254 Creaney J, Robinson BWS. Malignant mesothelioma biomarkers: from discovery to use in clinical practice for diagnosis, monitoring. Screening, and treatment. *Chest* 2017; 152: 143–149.

255 Woolhouse I, Bishop L, Darlison L, *et al.* BTS guideline for the investigation and management of malignant pleural mesothelioma. *BMJ Open Respir Res* 2018; 5: e000266.

256 Bibby AC, Maskell NA. Pleural biopsies in undiagnosed pleural effusions; Abrams *vs* image-guided *vs* thoracoscopic biopsies. *Curr Opin Pulm Med* 2016; 22: 392–398.

257 Mei F, Bonifazi M, Rota M, *et al.* Diagnostic yield and safety of image-guided pleural biopsy: a systematic review and meta-analysis. *Respiration* 2021; 100: 77–87.

258 Metintas M, Ak G, Metintas S, *et al.* Prospective study of the utility of computed tomography triage of pleural biopsy strategies in patients with pleural diseases. *J Bronchol Interv Pulmonol* 2019; 26: 210–218.

259 Dhooria S, Bal A, Sehgal IS, *et al.* Pleural cryobiopsy *versus* flexible forceps biopsy in subjects with undiagnosed exudative pleural effusions undergoing semirigid thoracoscopy: a crossover randomized trial (COFFEE Trial). *Respiration* 2019; 98: 133–141.

260 Janssen J, Maldonado F, Metintas M. What is the significance of non-specific pleuritis? A trick question. *Clin Respir J* 2018; 12: 2407–2410.

261 Pinelli V, Clive AO. Medical thoracoscopy in 2020: essential and future techniques. *In*: Maskell NA, Laursen CB, Lee YCG, *et al*, eds. Pleural Disease (ERS Monograph). Sheffield, European Respiratory Society, 2020; pp. 133–137.

262 Rozman A, Camlek L, Marc-Malovrh M, *et al.* Rigid *versus* semi-rigid thoracoscopy for the diagnosis of pleural disease: a randomized pilot study. *Respirology* 2013; 18: 704–710.

263 Dhooria S, Singh N, Aggarwal AN, *et al.* A randomized trial comparing the diagnostic yield of rigid and semirigid thoracoscopy in undiagnosed pleural effusions. *Respir Care* 2014; 59: 756–764.

264 Shafiq M, Sethi J, Ali MS, *et al.* Pleural cryobiopsy: a systematic review and meta-analysis. *Chest* 2019; 157: 223–230.

265 Giri M, Dai H, Guo S, *et al.* Efficacy and safety of pleural cryobiopsy *vs*. forceps biopsy for evaluation of undiagnosed pleural effusion: a systematic review and meta-analysis. *Front Med (Lausanne)* 2022; 9: 847146.

266 Wurps H, Schonfeld N, Bauer TT, *et al.* Intra-patient comparison of parietal pleural biopsies by rigid forceps, flexible forceps and cryoprobe obtained during medical thoracoscopy: a prospective series of 80 cases with pleural effusion. *BMC Pulmonary Medicine* 2016; 16: 98.

267 McDonald CM, Pierre C, de Perrot M, *et al.* Efficacy and cost of awake thoracoscopy and video-assisted thoracoscopic surgery in the undiagnosed pleural effusion. *Ann Thorac Surg* 2018; 106: 361–367.

Downloaded from https://publications.ersnet.org on August 13, 2026 at University of Liverpool. Please see licensing information on first page for reuse rights.